

WHEN DOES TRAINING ON DOWNSCALED IMAGES YIELD THE SAME GRADIENTS?

Seunghyun Ji

Independent researcher

standingbehindnv@gmail.com

ABSTRACT

Diffusion transformers deliver strong image generation, but their training cost grows superlinearly with resolution. A recent family of methods therefore trains or samples at reduced resolution on a spectral premise: at high noise, a downsampled latent preserves almost the full surviving signal. The evidence for that premise, however, lives at the level of generated output, which reads the substitution only through a full pipeline’s confounds; whether a downsampled step also delivers the *native training gradient direction*, under the same model and objective, has remained unresolved. We measure it directly, with a debiased gap estimator (naive finite-draw estimation manufactures the very effect under test), and propose a two-term account of what the substitution costs: a noise-dependent term governed by the downscale *ratio*, and a σ -independent floor governed by the target grid’s *absolute token count*, carried by the compute graph itself. On held-out routes it predicts where the spectral account, transported to the gradient and given its best tolerance, misses. On the mildest route (1024→896) the training gradient direction is indistinguishable from native at instrument resolution across the interior high-noise window $0.5 < \sigma \leq 0.94$, despite roughly a quarter fewer input tokens — a window no spectral tolerance predicts. Gating substitution on that window gives *selective* low-resolution LoRA training: -14.6% wall-clock at a fixed step budget, with a sample-level footprint within the training-seed lottery.

1 INTRODUCTION

The cost of a diffusion-transformer forward pass grows superlinearly with resolution: token count grows quadratically with edge length, and attention cost grows again in token count. That makes the title question worth asking precisely. At high noise, can the *gradient* of an adapter be computed on a downsampled latent as a drop-in substitute for the native one, with the same model, the same objective, and the same σ distribution, only the spatial grid changed on some steps?

There is a well-studied reason to hope the answer is yes. As the noise level σ grows, per-frequency signal-to-noise falls, and high spatial frequencies drop below the noise floor first, so that above a spectral crossover a downsampled latent carries *almost* the full surviving signal. But the studies of this resolution gap live almost entirely on the *inference* side: scale-wise distillation (Starodubcev et al., 2025), spectral progressive diffusion (Xiao et al., 2026), and spectrally-guided noise schedules (Esteves & Makadia, 2026) run high-noise *sampling* steps at reduced resolution and judge the premise by the generated output; and where resolution is reduced during training (Jin et al., 2024; Karras et al., 2018; Chen et al., 2024; Hoogeboom et al., 2023), the low-resolution stage is given its own, modified objective, and success is again read off the final samples. By the spectral premise, substitution is safe *whenever the noise has masked the frequencies the coarse grid loses*.

The gap at *training* time — whether a downsampled step delivers the native gradient direction under an unchanged objective — is not yet revealed unambiguously by any of this evidence: output-level success reads the substitution only through a full training or sampling pipeline’s confounds, and a small quality-score difference does not transfer to a small difference in what a training step *learns*. Measuring that gap directly is the route this paper takes. What a training step consumes is the expected adapter gradient, and *demotion* — our term for the training-time substitution of a downsampled, re-encoded latent for the native one — perturbs more than the network’s input: the regression target

carries the clean image at unit weight at every noise level, and the compute graph itself changes with the grid. Measuring it honestly is, unexpectedly, a metrology problem: the natural gap estimator (a redraw floor minus a single demoted-arm cosine) is biased by finite-draw variance, and the bias grows as the demoted grid shrinks, *manufacturing* exactly the token-count-ordered floor signature under test. Every claim below is stated in debiased units, at the instrument’s measured resolution.

The contributions:

1. **Two accounts, stated in the same units** (§3). We define the demotion gap at the gradient level, then derive in those units (a) the gradient gap the existing spectral argument implies — in its strongest tolerance-parameterized form (SPD, Xiao et al., 2026) — which is *ratio-only* and *floorless* by construction, and (b) our own account, which reads demotion as a perturbation of both factors of $g = J^T r$ and reduces, under four named assumptions, to a route amplitude on a universal mismatch curve *plus* a σ -independent graph floor. The two differ structurally before any measurement (§3.3), and every experiment is a place they disagree.
2. **A debiased measurement that scores both, and a predictive tool** (§4; the resulting (route, σ) map in §5.1). Measured with per-arm self-floors, a draw-count extrapolation, and a pre-specified margin whose certification power is set — not defined — by the instrument’s resolution, the mildest route (1024→896) is indistinguishable from the re-encode control over the interior high-noise window $\sigma \in (0.5, 0.94]$, while no noise level rescues a $2\times$ downscale. No member of the spectral tolerance family reproduces that map, and at the curve level its one free parameter is inert. Head to head on the same curves it misses at 0.147–0.355 RMSE where our account predicts held-out routes at ~ 0.07 – 0.09 — better than an oracle fit on the held-out data itself — with the floor law correctly predicting both held-out floors.
3. **Selective low-resolution training** (§5). Gating substitution on the measured-safe window gives an opt-in σ -conditional trainer realizing -15.1% forward FLOPs and -14.6% wall-clock at a fixed step budget, with a sample-level footprint within the training-seed lottery (Table 1); generation-quality non-inferiority at production scale remains the stated open gate.

2 RELATED WORK

Flow matching and DiT. Flow matching (Lipman et al., 2023) regresses a velocity field along prescribed probability paths between noise and data; rectified flow (Liu et al., 2023) takes the paths to be straight lines, giving the linear noising $z_\sigma = (1 - \sigma)x + \sigma\epsilon$ and the constant target $v = \epsilon - x$ used throughout — the training objective of current large text-to-image systems (Esser et al., 2024). The Diffusion Transformer (Peebles & Xie, 2023) replaces the U-Net denoiser with a transformer over patchified latent tokens, so a spatial grid enters the network only as a token sequence: token count grows quadratically with edge length and attention cost quadratically again in token count, the cost structure that makes resolution the expensive axis. Spatial position is now widely carried by a grid-dependent coordinate system, most commonly rotary embeddings (Su et al., 2024).

Scale-wise and progressive-resolution diffusion. Progressive growing (Karras et al., 2018) and resolution curricula (Chen et al., 2024; Hoogetboom et al., 2023) train at increasing resolution as a schedule, each low-resolution stage carrying its own stage-specific objective. A more recent family ties resolution to the noise level itself, on the spectral premise above, and has so far been studied on the *inference* side. Scale-wise distillation (SwD) (Starodubcev et al., 2025) states the claim directly (above the crossover, the noised downsampled latent is close in distribution to the downsampled noised latent) and samples early steps at reduced resolution. Spectral Progressive Diffusion (SPD) (Xiao et al., 2026) gives the argument its sharpest available form: a tolerance-parameterized per-frequency activation time under a diagonal Gaussian spectral model — a criterion on the *Bayes-optimal predictor*’s per-band output rather than merely on input SNR — from which an inference-time resolution schedule follows. Spectrally-guided noise schedules (Esteves & Makadia, 2026) apply the same reasoning to schedule design; and where the premise does enter training, as in pyramidal flow matching (Jin et al., 2024), the objective is restaged per pyramid level rather than left native.

Positional extension. Position interpolation (Chen et al., 2023), banded/frequency-selective variants (YaRN, Peng et al., 2024), and diffusion-specific dynamic position extrapolation (Issachar et al., 2026; Zhao et al., 2026) rescale rotary coordinates to evaluate a model at unseen grid sizes. We use exact positional interpolation as a *causal intervention* (matching the downscaled grid’s relative phase geometry to native) to decompose the resolution-sensitivity floor, and adopt the σ -gated band schedule of Zhao et al. (2026) as a training-time refinement.

3 TWO ACCOUNTS OF WHAT DEMOTION COSTS

This section fixes the estimand (§3.1), then states two accounts of it in those same units: a training-gap interpretation built on the spectral account of the inference-side literature (§3.2), and ours (§3.3). They are not variants of one model: they disagree about which quantity governs a route’s cost, about what happens at pure noise, and about whether a single parameter can order all routes (§3.3).

3.1 THE ESTIMAND

Objective and noising. A flow-matching model regresses the constant-velocity transport between a clean latent and pure noise. With clean latent x , noise $\epsilon \sim \mathcal{N}(0, I)$, and noise level $\sigma \in (0, 1]$, the noised input is $z_\sigma = (1 - \sigma)x + \sigma\epsilon$ and the target is the velocity $v = \epsilon - x$; with caption condition c the loss is $\|\hat{v}_\theta(z_\sigma, \sigma, c) - v\|^2$.

Demotion. We write $e_0 \rightarrow e$ for a *demotion*: the native latent at the route’s source edge e_0 is replaced with a coarser-grid one at its target edge e . With y the image at its native resolution, $\mathcal{E}$ the VAE encoder, and $\mathcal{R}_e$ the pixel-space downscale to the tier- e bucket (native aspect ratio preserved), $x_{\text{src}} = \mathcal{E}(y) \rightarrow x_{\text{dem}} = \mathcal{E}(\mathcal{R}_e(y))$ — a latent with proportionally fewer spatial tokens, the downscale acting in pixel space, not latent space. Everything else is unchanged: the caption, the σ under test, and the noise distribution (ϵ drawn at the tier- e shape). The demoted step trains on $z_\sigma = (1 - \sigma)x_{\text{dem}} + \sigma\epsilon$ with target $v = \epsilon - x_{\text{dem}}$, so the substitution changes both what the network sees and what it is asked to predict, but only through the spatial grid.

The estimand. Three *arms* recur throughout: the source arm src , which trains on the native-resolution latent x_{src} ; the demoted arm dem , which trains on x_{dem} ; and the control arm reenc , which decodes and re-encodes the native latent at native resolution: it pays the VAE round trip that demotion necessarily pays, with no grid change. For an arm $a \in \{\text{src}, \text{reenc}, \text{dem}\}$, let $\bar{g}_a(\sigma) = \mathbb{E}_\epsilon [\nabla_\theta \mathcal{L}]$ be the population per-bin mean adapter gradient of a query (image, caption). The *population demotion distance* of the route, $d_{e_0 \rightarrow e}$, is

$$d_{e_0 \rightarrow e}(\sigma) = 1 - \cos(\bar{g}_{\text{src}}(\sigma), \bar{g}_{\text{dem}}(\sigma)) = \frac{1}{2} \|\hat{g}_{\text{src}} - \hat{g}_{\text{dem}}\|^2 \geq 0, \quad (1)$$

with $\hat{g}$ the unit-normalized gradients. Route-level scalars carry the *route* as a subscript, abbreviated to the target edge alone (d_e) wherever the source is fixed by context; the control’s distance d_{reenc} carries its arm label instead, since it changes no grid (full convention: Appendix A). The *reported* quantity of every bin-resolved map and every fit in this paper is the *re-encoding excess*

$$\text{gap}_e(\sigma) := d_e(\sigma) - d_{\text{reenc}}(\sigma), \quad (2)$$

the demotion distance beyond what the control already pays; gap_e can be negative, d_e cannot. End-point and floor tables report debiased estimates of d_e with the control as its own row.

The two aggregations. In a batched training scenario Eq. 1 admits two aggregations over a probe set: per-example (cosine per image, then mean) or batch-aggregate (gradients summed over the batch before the cosine, the object SGD actually follows). Writing each image’s demotion-induced gradient disagreement as a coherent mean b plus a zero-mean idiosyncratic deviation with covariance Σ_η , the batch-aggregate distance at batch size B is, to second order,

$$\mathbb{E}[d_B] \approx \frac{\|P^\perp b\|^2}{2\|\mu\|^2} + \frac{\text{tr}(P^\perp \Sigma_\eta P^\perp)}{2B\|\mu\|^2},$$

with μ the mean source gradient and $P^\perp$ the projector off $\hat{\mu}$: an intercept (the coherent share, which never averages out) plus a $1/B$ term. The per-example map is therefore *not* a lower bound for larger

batches — monotone improvement with B would require an iid zero-mean disagreement model we have not verified — and no coefficient fitted on one is guaranteed to transfer to the other. These are different estimands; the aggregation operator is part of the estimand, and §5.1 publishes one map for each.

3.2 THE SPECTRAL ACCOUNT

The spectral argument of §2 supplies the field’s working model of the same gap, and it is worth deriving in the units Eq. 1 just fixed. Its strongest available form is the tolerance-parameterized crossover of Xiao et al. (2026).

Demotion deletes a band. Write $u^{(\omega)}$ for the frequency- ω component of a field u in the latent’s spatial Fourier decomposition, with ω in normalized units — cycles per sample of the *native* grid, so the native Nyquist sits at $\frac{1}{2}$. The downscale $\mathcal{R}_e$ resamples the image with e/e_0 times fewer samples per axis, and a grid at that density can represent no frequency above half its own sampling rate; the resampler’s anti-aliasing prefilter removes, rather than aliases, whatever lies above. Idealizing the VAE round trip away — its residue is exactly what the reenc control of §3.1 pays — demotion is an ideal low-pass:

$$x_{\text{dem}}^{(\omega)} = \begin{cases} x^{(\omega)}, & |\omega| < \omega_e, \\ 0, & |\omega| \geq \omega_e, \end{cases} \quad \omega_e = \frac{1}{2} \frac{e}{e_0}, \quad (3)$$

deleting exactly the bands above the coarse grid’s Nyquist frequency ω_e and leaving the rest intact.

When a band stops mattering. The noising is linear, so it acts band by band: $z_\sigma^{(\omega)} = (1-\sigma)x^{(\omega)} + \sigma \epsilon^{(\omega)}$, and the velocity target splits as $v^{(\omega)} = \epsilon^{(\omega)} - x^{(\omega)}$. Under the premise that each clean-latent band is independently Gaussian, $x^{(\omega)} \sim \mathcal{N}(0, P_\omega)$ with P the clean-latent power spectrum, the *Bayes-optimal* per-band velocity prediction $v^*(\omega) = \mathbb{E}[v^{(\omega)} | z_\sigma]$ is a per-band linear shrinkage of the input, and as σ grows the band’s posterior collapses to its prior: the optimal prediction tends to the pure-noise answer $\epsilon^{(\omega)}$. The crossover t_ω makes this quantitative — for all $\sigma \geq t_\omega$ the prediction is within a tolerance δ of pure noise, $\mathbb{E}[|v^*(\omega) - \epsilon^{(\omega)}|^2] \leq \delta$. The tolerance is the account’s one free parameter, setting how strict “stops mattering” is; choosing $\delta = (1+P_\omega)/2$ recovers the often-quoted equal-power (input-SNR) crossover $\sigma_{\text{eq}} = \sqrt{P_\omega}/(1 + \sqrt{P_\omega})$ as one slice of the family. How δ is set in practice — including a zero-free-parameter anchor measured from our own pipeline — is treated in Appendix G; no conclusion below hinges on the choice.

Carried to the training gradient. The destroyed bands of Eq. 3 all sit at or above ω_e ; on a decaying spectrum the lowest of them is the most powerful, so it crosses last, and the safe-substitution boundary for the whole route sits at t_{ω_e} . Under the premise that demotion costs only what the destroyed bands still contribute to the prediction, the cost has one place to land: whatever those bands contribute to the residual, and through the backward pass to the adapter gradient, perturbs $\bar{g}_{\text{dem}}$ away from $\bar{g}_{\text{src}}$. Writing $S_e(\sigma)$ for that destroyed-band contribution to the distance of Eq. 1, the implied gradient gap is S_e gated at the Nyquist band’s activation time:

$$\text{gap}_e^{\text{spec}}(\sigma) = S_e(\sigma) \mathbf{1}[\sigma < t_{\omega_e}], \quad t_\omega = \frac{1}{1 + \sqrt{\delta / (P_\omega (1 + P_\omega - \delta))}}. \quad (4)$$

Under the same premise the account also fixes the residual level: with no cross-frequency coupling, the cross-grid mean-residual mismatch is confined to the destroyed band and ordered across routes by destroyed-band energy (the Gaussian posterior mean is the classical per-band Wiener shrinkage, Wiener, 1949).

Three structural consequences are fixed by the *form* of Eq. 4, before any gradient is measured, and each is a place the account can be falsified without arguing about δ . **(a) One parameter orders every route:** all boundaries move together under the single tolerance, so the family is totally ordered. **(b) Ratio only:** the cut $\omega_e = \frac{1}{2} e/e_0$ of Eq. 3 is a function of the downscale *ratio* alone, so the account is structurally blind to any absolute-size effect. **(c) No floor:** for every route and every δ the predicted gap vanishes above a finite boundary, so no σ -independent term is expressible.

3.3 OUR ACCOUNT: WHAT DEMOTION PERTURBS, AND ITS ANGULAR REDUCTION

We start one level lower, from what the substitution touches in the gradient itself. Differentiating the loss of §3.1 gives, per draw and up to a constant, $g = \nabla_{\theta} \frac{1}{2} \|\hat{v}_{\theta} - v\|^2 = J^T r$, with $r = \hat{v}_{\theta} - v$ the prediction residual and $J = \partial \hat{v}_{\theta} / \partial \theta$ the Jacobian through which the compute graph turns a residual into an adapter gradient. Demotion changes exactly the two factors of this product, and nothing else about the map from (image, noise, caption) to adapter gradient: the residual, because it changes the data the objective consumes, and the Jacobian, because it changes the compute graph the gradient flows through. Averaging over draws and expanding the product (Appendix B), the perturbation $\delta g = \bar{g}_{\text{dem}} - \bar{g}_{\text{src}}$ decomposes as

$$\delta g = \underbrace{J^T \Delta \bar{r}(\sigma)}_{\text{data branch } B_e} + \underbrace{\Delta J^T \bar{r}}_{\text{graph branch } C_e} + \underbrace{\Delta J^T \Delta \bar{r} + \dots}_{\text{remainder } R_e}. \quad (5)$$

J_{src} and J_{dem} act on different grids, so the branches are not literal matrix products across arms; they are defined *operationally*, as the two independent perturbation directions in the shared adapter-parameter space where both gradients live, and §4.2 isolates each by its own intervention.

Read against Eq. 5, the spectral account of §3.2 is a *strict special case*: it keeps the input-mediated part of the data branch, gated at Nyquist, and sets everything else to zero — no target-mediated share, no graph branch ($\Delta J \equiv 0$), no interaction. Both omissions are structural. The graph branch does not reference σ at all, since the grid change does not: token count, rotary phase density, and sequence-length-dependent normalization change identically at every noise level, so nothing about noise masking can remove them. And within the data branch, only the *input*-mediated part shrinks with σ : perturbing the image by δx perturbs the residual by $D_z f[(1-\sigma)\delta x] + \delta x$, and the second, target-mediated term carries unit weight at every σ . At $\sigma=1$ the input is pure noise on every grid, so the input-mediated part vanishes in distribution, but the endpoint is not data-free by construction; whether the surviving target-mediated share is *resolvable in the angular estimand* is an empirical question, discharged in §4.2. Input-level spectral sufficiency licenses inference-time substitution; what training consumes is $\mathbb{E}_{\epsilon}[\nabla_{\theta} \mathcal{L}]$, and that is not bounded by an input-masking argument.

The residual factor has an exact posterior form. The data branch’s σ -curve ingredient $\Delta \bar{r}$ is not a black box. For the population-optimal (Bayes) velocity field $v_e^*(z, \sigma, c) = \mathbb{E}[v \mid z, c] = (z - m_e(z, c))/\sigma$, with $m_e(z, c) = \mathbb{E}[x \mid z, c]$ the grid- e posterior mean, the fixed-image mean residual evaluates exactly to

$$\bar{r}_e^*(\sigma; x) = \frac{x - \mathbb{E}_{z, \sigma | x}[m_e(z, \sigma, c)]}{\sigma}, \quad \bar{r}_e^*(1; x) = x - \mathbb{E}[x \mid c], \quad (6)$$

so the cross-grid mismatch is a difference of two posterior reconstruction-bias operators, and the target-mediated $\sigma=1$ survivor of the previous paragraph is exactly the image’s deviation from the caption-conditional prior mean (derivation, score form, and a Gaussian closure: Appendix C). Two consequences are used below. First, MSE optimality zeroes the residual conditional on the *input*, not the image: $\mathbb{E}[v^* - v \mid z, c] = 0$ does not imply $\mathbb{E}[v^* - v \mid x, c] = 0$, and Eq. 6 is generally nonzero — so the measured residual of a trained network splits exactly as $\bar{r}_{\theta, e} = \bar{r}_e^* + \mathbb{E}_{z, \sigma | x}[q_{\theta, e}]$, intrinsic posterior bias plus approximation error ($q_{\theta, e} = \hat{v}_{\theta, e} - v_e^*$), a *mean-residual* mismatch rather than a model-error difference alone. Second, the noising path fixes only how the posterior operators are smoothed in σ ; a numerical curve requires a model of the grid-conditional clean-latent distribution, which is not analytically available. In these terms the spectral account’s residual-level form (§3.2) is the *diagonal-Gaussian* closure of this identity, and the measured $\|\Delta \bar{r}(\sigma)\|$ consumed by assumption (iv) below is its *empirical* closure — an estimate of a defined functional of the two posterior operators, not an unconstrained fitted shape.

It remains to read Eq. 5 in the units of the estimand (step-by-step in Appendix D). Writing $u^{\perp} = u - (\hat{g}_{\text{src}}^T u) \hat{g}_{\text{src}}$ for the component orthogonal to $\bar{g}_{\text{src}}(\sigma)$, split the perturbation into a rescaling and a rotation: $\kappa_{\parallel} = \hat{g}_{\text{src}}^T \delta g / \|\bar{g}_{\text{src}}\|$ and $\kappa_{\perp} = \|\delta g^{\perp}\| / \|\bar{g}_{\text{src}}\|$. Substituting $\bar{g}_{\text{dem}} = \bar{g}_{\text{src}} + \delta g$ into Eq. 1 gives, *exactly*,

$$d_e = 1 - \frac{1}{\sqrt{1 + \kappa_{\text{eff}}^2}}, \quad \kappa_{\text{eff}} = \frac{\kappa_{\perp}}{1 + \kappa_{\parallel}} : \quad (7)$$

the gap is a saturating function of a single rotation-to-scale ratio — only the orthogonal component rotates the gradient direction, and a parallel component merely renormalizes it. Taylor-expanding Eq. 7 at small κ and substituting Eq. 5 into the square gives the *four-term angular expansion*, derived as unconditionally as Eq. 7 itself:

$$d_e(\sigma) \approx \underbrace{\frac{\|B_e^\perp\|^2}{2 \|\bar{g}_{\text{src}}\|^2}}_{\text{data share } S_e} + \underbrace{\frac{\|C_e^\perp\|^2}{2 \|\bar{g}_{\text{src}}\|^2}}_{\text{graph share } F_e} + \underbrace{\frac{\langle B_e^\perp, C_e^\perp \rangle}{\|\bar{g}_{\text{src}}\|^2}}_{\text{projected interaction } I_e} + R'_e, \quad (8)$$

where R'_e collects the remainder branch, the beyond-quadratic terms of Eq. 7, and the parallel-component correction (exact content: Appendix D). S_e and F_e are non-negative; I_e can carry either sign, a fact that matters in Appendix F. At the harshest measured operating point ($d = 0.30$ at 512, $\sigma=1$) the implied perturbation is $\kappa_{\text{eff}} \approx 1.02$ — not small — so quadratic-order reads on that route are indicative and the exact form carries the quantitative claims. Note that Eq. 4 is Eq. 8 with $F_e = I_e = 0$ and S_e gated at Nyquist.

From the expansion to a measurable form. The reported object is the excess over the control, whose own expansion is degenerate — the reenc arm changes no grid, so $\Delta J_{\text{reenc}} = 0$ and both terms built from it vanish, leaving $d_{\text{reenc}} \approx S_{\text{reenc}} + R'_{\text{reenc}}$. Subtracting term by term therefore cancels only the re-encode component of the data share and passes the graph share and interaction through untouched. Four named assumptions, each tested by a designated probe in §4, then trade the exact book-keeping (in which no term is separately measurable per bin) for a form built from measurable curves and per-route constants: (i) **small remainder**, R'_e negligible over the claimed domain; (ii) **negligible projected interaction**, $|I_e| \ll S_e + F_e$; (iii) **graph-relative stationarity**, $\|C_e^\perp\|/\|\bar{g}_{\text{src}}\|$ σ -constant (numerator and denominator share the residual’s scale), so F_e is flat in σ — write Floor_e for its value; (iv) **data-branch factorization**, $\|B_e^\perp\| \approx a_e \|\Delta\bar{r}(\sigma)\|$: a σ -independent route amplitude times the *norm* of the cross-grid mean-prediction-residual mismatch of Eq. 5, asserted to be route-independent and directly measurable as such (§4.2). The first two are not unstructured hopes: both discarded terms obey exact Cauchy–Schwarz bounds in the retained shares ($|I_e| \leq 2\sqrt{S_e F_e} \leq S_e + F_e$, the remainder entering at half power in its own share), so (ii) carries independent content only near share crossover (Appendix D). Applied,

$$\text{gap}_e(\sigma) \approx \underbrace{\frac{1}{2} a_e^2 \cdot (\|\Delta\bar{r}(\sigma)\| / \|\bar{g}_{\text{src}}(\sigma)\|)^2}_{\text{data term}} + \underbrace{\text{Floor}_e}_{\text{graph term}}, \quad (9)$$

the *two-term account*, stated on the reported excess. The quadratic power is fixed by the cosine geometry *locally*; far from the small-mismatch regime the licensed read is the un-expanded parent form, Eq. 7 with loading $\kappa_e(\sigma) = a_e \|\Delta\bar{r}(\sigma)\|/\|\bar{g}_{\text{src}}(\sigma)\|$ (the same amplitude, recovering Eq. 9’s data term in the small- κ limit), whose data term *saturates* instead of overshooting. The loading itself is an *empirical* ingredient, not a derivation, so Appendix F scores both forms held out, labelling Eq. 7 the best empirical predictor and Eq. 9 the derived small-perturbation form. To keep the ledger explicit: Eqs. 7 and 8 are unconditional; Eq. 9 is conditional on the four assumptions; the coefficients a_e , $\|\Delta\bar{r}(\sigma)\|$, $\|\bar{g}_{\text{src}}(\sigma)\|$ and Floor_e are measured. The account was pre-registered with an outcome matrix — including a kill switch, all endpoint gaps ≈ 0 , that would have falsified the graph term — before the discriminating runs.

The graph term has identifiable parts. Changing the grid changes (i) the rotary coordinate system, in which every band of the position embedding accumulates phase at a different density across the image, and (ii) non-positional graph statistics: attention softmax over a different token count, sequence-length-dependent normalization, and the capacity of the coarse graph to approximate the fine graph’s computation (the conceptual frame is attention as a discretization of a function-space operator, Calvello et al., 2024, whose statistics need not be discretization-invariant). Splitting $\text{Floor}_e = \text{RoPE}_e + \text{Resid}_e$ yields one sharp prediction: the positional share is a *coordinate-system artifact*, so an exact positional-interpolation intervention should remove RoPE_e wherever the data branch is inactive ($\sigma \rightarrow 1$); a band-counting sketch attaches an *absolute-size* governor to Resid_e . Like YaRN’s own band thresholds (Peng et al., 2024), that sketch’s constants are calibrated, not derived; we claim no closed form for Floor_e .

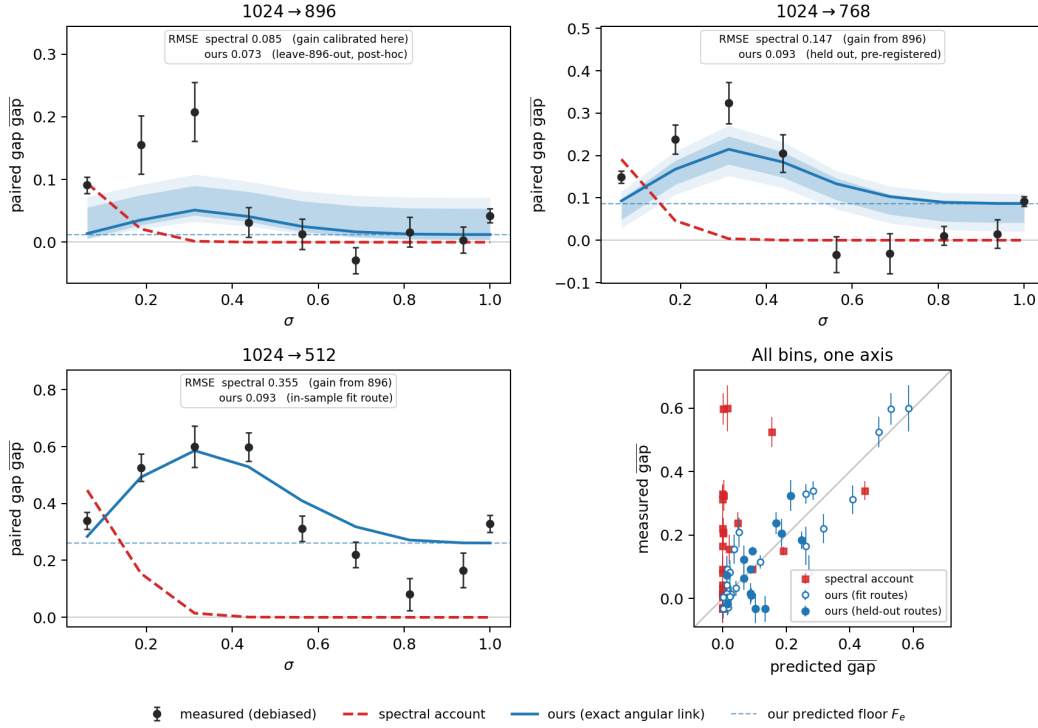


Figure 1: **Both accounts on the same axes.** Per route, one set of measured debiased gap curves (Eq. 1, black) carries both predictions of §3: the spectral account (red dashed) and ours under the exact angular link (blue, Eq. 7; three-form comparison in Appendix Fig. 3), with each route’s predicted floor (blue dashed — dashed-to-solid is the data term) and a 68%/95% full-pipeline bootstrap band ($B=1000$; Appendix E). Neither account is fit to the route it is drawn on except as labelled in-panel: ours predicts 1024→768 from governors alone with zero per-route freedom (pre-registered held-out) and 1024→896 under the post-hoc leave-896-out re-derivation, while the spectral account’s single gain is calibrated on 1024→896 and frozen. Unit honesty: the red curve is *the spectral account read through our bridge*, since it emits residual units only, with the tolerance δ set to the measured anchor δ_{reenc} ; the transported curve dies by $\sigma \approx 0.35$ on every route, so every tolerance from the equal-power slice to δ_{reenc} scores within 0.01 RMSE of every other — δ is inert at the curve level (the whole family, and the boundary sweep, in Appendix Fig. 5). The 768 mid- σ dip sits outside the 95% band and is our reduction’s one structural failure (Appendix F). Bottom right: predicted vs measured over every bin, with ours also on the two 1280-tier routes the spectral bridge does not cover.

The two accounts, side by side. Both accounts agree that gaps shrink with σ and order by severity. Everything else is fixed by the *form* of Eq. 4 against Eq. 9 before any gradient is measured: what orders routes (the downscale ratio alone, against the ratio through the amplitude a_e and the absolute target token count through the floor Floor_e); what is free (one tolerance δ shared by every route, against per-route a_e, Floor_e generated by two governors); the endpoint ($\text{gap}_e \rightarrow 0$ on every route, against $\text{gap}_e \rightarrow \text{Floor}_e > 0$); the curve shape (a gated step at t_{ω_e} , against $\|\Delta\bar{F}(\sigma)\|/\|\bar{g}_{\text{src}}(\sigma)\|$ carried through Eq. 7); the cross-grid residual $\Delta\bar{F}$ (confined to the destroyed band and ordered by its energy, against a norm shared by all routes with route identity living in J); and equal-ratio routes at different absolute sizes (identical, against same a_e and different Floor_e). Each disagreement is an experiment in §4, and two are decidable without fitting anything: if a route’s gap survives at $\sigma = 1$, where the input is pure noise on every grid, no member of the spectral family can produce it; and if two routes of the *same ratio* at different absolute sizes differ, ratio is not the whole governor.

4 EXPERIMENTS: BOTH ACCOUNTS, SCORED

Figure 1 states the comparison before the apparatus: the same measured, debiased gap curves with each account of §3 laid over them. The existing account is near zero where the measurement peaks and predicts zero where the measurement floors; ours tracks both, on routes it never saw. The rest of this section earns those overlays. We build the instrument (§4.1) and score both accounts against the measured curves — the spectral one at the boundary and the curve level, ours term by term (§4.2); the per-route map those curves certify is read in §5.1, where it becomes the deployed artifact, and the held-out prediction protocol behind the figure’s overlays, its pre-registered gates, and the one structural failure it exposes are quantified in Appendix F. Each subsection states the prediction it discharges against criteria frozen before the runs; several probes double as falsifiers of our own account’s assumptions.

4.1 THE INSTRUMENT

Model and data. All measurements use Anima (CircleStone Labs & Comfy Org, 2025), an open DiT-based flow-matching text-to-image model (2B parameters, 28 transformer blocks, 3D rotary position embeddings, and the Qwen-Image VAE, Wu et al., 2025), fine-tuned with LoRA adapters on an illustration corpus. Images are bucketed by native aspect ratio into resolution *tiers* indexed by nominal edge length $e \in \{512, 768, 896, 1024, 1280\}$; a tier- e image occupies roughly $(e/16)^2$ latent-patch tokens ($\sim 4,100$ at 1024, $\sim 3,000$ at 896, $\sim 2,160$ at 768, $\sim 1,000$ at 512). The probe adapter is a plain LoRA checkpoint trained at native tiers; we return to this operating-point dependence in §6.

The gradient probe. For each image and each of B uniform σ -bins we accumulate adapter gradients over D stratified noise draws per arm¹ (verdict runs: 8×8 ; $N = 40$ images) and compare arms by cosine similarity of the flattened accumulated gradients: a **floor** (two independent draw sets at native resolution, the redraw null); the **reenc** control of §3.1; one **demote** arm per edge e ; and, on debiased runs, per-arm **self-floors** (a second independent draw set for every arm). The gap is in cosine units, a scale-free mismatch *fraction* (bin-mean SEM ~ 0.02 at $N=40$). Gap subtraction, not absolute cosines, is the valid read: the floor itself moves with $\|g\|$ across bins. Every bin-mean curve must pass a split-half reliability check over images, a discipline imposed after an earlier failure in which per-image rankings from the same gradients had reliability indistinguishable from zero.

The debiased estimator. Two corrections, used together, remove the finite-draw attenuation. First, *self-floors with attenuation correction*: every arm runs a second independent draw set g'_a , giving a per-arm self-cosine $\cos_{\text{self}} = \cos(g_a, g'_a)$, so that each arm has the same two-estimate structure the native floor has. The debiased cosine is the classical split-half disattenuation correction,

$$\hat{c} = \frac{\cos(\bar{g}_{\text{src}}, \bar{g}_a)}{\sqrt{\cos_{\text{floor}} \cdot \cos_{\text{self},a}}}, \quad \widehat{\text{gap}} = 1 - \hat{c}, \quad (10)$$

which cancels the finite-draw attenuation of numerator and denominator to first order. Second, *draw-count extrapolation*: the attenuation of an accumulated-draw cosine decays as $1/D$, so sweeping nested draw subsets and fitting $\text{gap}(D) = \text{gap}_\infty + c/D$ reads off the draw-limit gap directly. The two are mutually checking, and the check passes: debiased fits come out D -flat, and the nested sweep extrapolates the native redraw floor to a self-cosine of ≈ 1 — the population per-bin native gradient is a *fixed direction*, and the entire redraw floor is finite-draw noise (validation numbers: Appendix E). No conclusion in this paper survives on the naive estimator alone.

¹Finite draws are not innocuous here. Every finite-draw cosine is attenuated below its population value; the attenuation is arm-dependent, because per-draw gradient variance grows as the token count falls (the MSE loss averages over fewer elements); and the natural estimate of Eq. 1 — a redraw floor minus a single demoted-arm cosine — is asymmetric, the floor comparing *two* finite-draw estimates of the native gradient where each demote arm contributes one. An iso-direction null (demotion changing *nothing* but the variance) therefore still produces a positive gap that *grows as the target grid shrinks* — the same signature as “absolute token count sets the floor” — and no naive read can tell the two apart. The mechanism is live and large at our operating point (measured magnitudes and c/D decay: Appendix E).

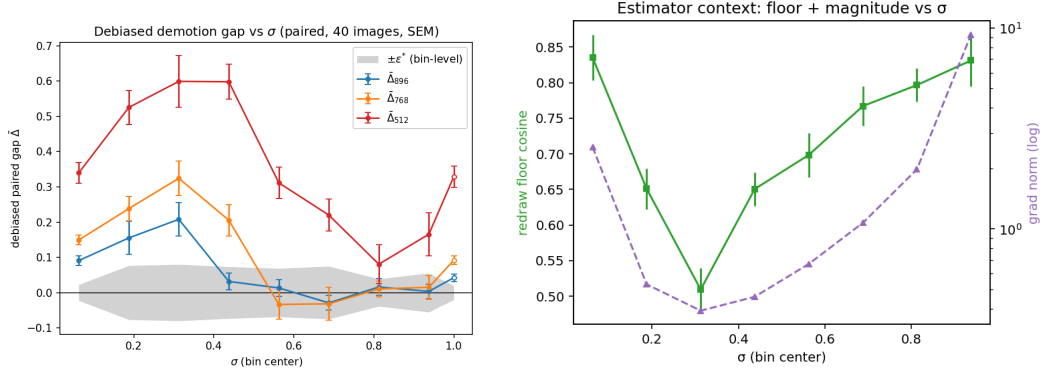


Figure 2: The measurement in detail. (a) Debiased demotion gap per σ -bin, paired against the re-encoding control (1024-tier natives, $N=40$; per-bin values and SEMs in Appendix Table 6; gray band: bin-level $\pm\epsilon^*$ (§4.1); open markers: the $\sigma=1$ endpoint bin): safety is a per-route boundary — the 896 route enters the band at $\sigma \approx 0.5$, 768 has no certifiable window, and 512 never falls inside it (grazing its boundary at $\sigma = 0.81$) — against the spectral family’s prediction of $\sigma \approx 0.14\text{--}0.20$. (b) The estimator context those curves are read through: the redraw-floor cosine and the U-shaped gradient norm $\|\bar{g}_{\text{src}}(\sigma)\|$.

Margins, resolution, pairing, and coverage. Any estimate of $\text{gap}_e(\sigma)$ compares finite-draw estimates of population directions, so a safety verdict needs two numbers, and they must be stated *separately*: a margin — the largest excess considered practically equivalent, a property of the question — and the instrument’s resolution, which determines only the power available to certify that margin. Conflating them would let a noisier experiment buy itself a looser definition of safety. We fix the practical margin at $\epsilon = 0.02$, comparable to the re-encode control’s own confidence half-width (Table 7), independent of N , D , and the observed standard errors. A route is *certified safe at margin ϵ* in a bin exactly where the one-sided non-inferiority bound clears it,

$$\text{UB}_{95}(\overline{\text{gap}}) < \epsilon, \quad (11)$$

and a *window* claim over several bins requires the simultaneous (Bonferroni-corrected) one-sided bounds to clear it in every bin. Resolution enters only through power: for a route whose true excess is zero the paired estimate $\overline{\text{gap}}$ fluctuates with standard error $\text{SE}(N, D, \sigma\text{-bin})$, and its zero-excess upper bound $\epsilon^*(N, D, \text{bin}) = 1.645 \text{ SE}$ is the *certification resolution*. A cell with $\epsilon^* > \epsilon$ is *underpowered* at the margin: a crossing there can occur on a favorable draw and is reported but not counted as certification, and the remedy is more images or draws, never a looser margin. “No certifiable window” means a route’s simultaneous upper bounds do not cross below ϵ at the current (N, D) ; “certifiably unsafe” means the corresponding lower bound stays above it. At the verdict grid’s operating point ($N=40$, $D=8/\text{bin}$) the bin-level ϵ^* is 0.03–0.08 — underpowered at $\epsilon = 0.02$, so interior verdicts below are stated as the margins they *do* certify — while endpoint-mode draw sweeps (D up to 64) reach $\epsilon^* \approx 0.01\text{--}0.02$ and are powered for the strict margin. At $D=8$ the *unpaired* debiased estimator is unstable where the floors are small (mid- σ ; the denominator is noisy), so all verdict maps report the *paired per-image excess* $\text{gap}_{e,i} = \widehat{\text{gap}}_{e,i} - \widehat{\text{gap}}_{\text{reenc},i}$, trimmed of non-finite and $|\text{gap}_{e,i}| > 1.5$ values; pairing cancels the shared per-image, per-bin draw structure. Finite-draw cosines are also compiler-kernel-path sensitive, so all pairings live within one run. Every number in the main text is debiased; four secondary probes (the 1280-tier iso-severity twin, the 896-tier transfer, the depth split, and the positional-interpolation intervention) predate the debiased instrument and are retained only where the strictly positive bias makes an in-band read conservative, or where a paired same-grid contrast cancels the bias to first order (Appendices E, I).

4.2 BOTH ACCOUNTS, SCORED

Both accounts are scored against one measured object: the debiased gap curves of Figs. 1 and 2, whose per-bin reading — and the two revisions debiasing forced on the historical record — §5.1

sets out in full. Four features of those curves are all the scoring needs: the 896 route enters the re-encoding band at $\sigma \approx 0.5$, 768 has no certifiable window, 512 is gapped at every noise level, and every route’s high- σ plateau converges to a nonzero level ordered by the target grid’s token count. We score the spectral account against those features first, at the boundary and then the curve level, and ours after, term by term.

The spectral account: boundaries, curves, and an inert δ . The Qwen-Image VAE’s latents are high-frequency-quiet ($P(f) < 1$ above $f \approx 0.16$ cycles/latent-pixel; latent variance 0.42), giving equal-power crossovers of 0.136/0.146/ ≈ 0.20 for 896/768/512 with tight per-image spread (p10–p90 ≈ 0.11 –0.16): at that tolerance demotion would be safe over almost the entire σ axis. Table 5 (Appendix G) sweeps the whole family, and no row reproduces the measured map (§5.1): the δ matching the measured 896 boundary (0.025) predicts 768 safe at 0.52 and 512 safe at 0.62, while $\delta = 0.01$, the published default of Xiao et al. (2026), predicts all three safe by 0.72. Appendix G also closes the “wrong δ ” objection by construction: our own pipeline’s measured re-encode error anchors the family with zero free parameters ($\delta_{\text{reenc}} \approx 10^{-5}$), and the anchored boundaries, measured where they move, land together at $t^* \approx 0.98$ — maximally conservative and still structurally wrong in both directions at once. At the curve level (Fig. 5, the family transported to full curves) the same failure is sharper: the latents are high-frequency-quiet, so the predicted mismatch is concentrated below $\sigma \approx 0.3$ and the transported curve is near zero exactly where the measured curves peak (RMSE 0.147 at 768 and 0.355 at 512), while above each tolerance’s committed boundary, where every member predicts exactly zero, the floor persists (committed-region RMSE 0.027/0.049/0.218 for 896/768/512 at SPD’s default). And because the transported curve has already died by $\sigma \approx 0.35$, every tolerance from the equal-power slice to the measured anchor scores within 0.01 RMSE of every other: δ , the family’s one degree of freedom, is *inert* at the curve level, so the structural failures of §3.2(a–c) do not hinge on its choice. Both reads are insensitive to the residual-shape choice in the transport (per-draw vs mean-residual within 10%; identical committed-region numbers).

Ours, first term: the graph term exists (three probes, one number). At the $\sigma = 1$ *endpoint* the input is pure ϵ , identical in distribution across arms, so the input-mediated data term is zero by construction; the debiased endpoint floors are +0.019/+0.056/+0.304 for 896/768/512 (Appendix Table 7), and the 512 value clears the pre-registered confirmation bar ($\widehat{\text{gap}}_\infty \geq 0.15$) on 12 of 12 probe images individually. Per §3.3 the target-mediated share need not vanish there, so two further probes close the step. The *x-zero* probe removes the image from input *and* target everywhere, isolating pure graph-shape sensitivity, and its debiased draw-limit gaps equal the endpoint gaps at every route. The *target-strength sweep* re-runs the endpoint with target $\epsilon - \alpha x$, $\alpha \in \{0, \dots, 1\}$, and the paired debiased excess is α -flat at the well-conditioned anchors (e.g. 768: $+0.070 \pm 0.015$ at $\alpha=0$ vs $+0.049 \pm 0.010$ at $\alpha=1$; control slope $+0.003$): no resolvable target-content share *in the angular estimand*. The flatness is not a cancellation mystery: the target-mediated gradient itself is real and large, but demotion’s change to it lands almost entirely *along* $\hat{g}_{\text{src}}$, to which the cosine gap is blind (Eq. 7) — the endpoint gap is the graph floor as the angular estimand reads it (parallel-landing decomposition, with the $\kappa_{\parallel}/\kappa_{\perp}$ numbers, in Appendix E). Two exonerations complete the picture. The re-encoding control sits at zero within CI, so the VAE round trip is not the cause — and, conversely, that round trip has *measurable* input-level error at below-resolution gradient cost, so input-level error neither implies nor excuses gradient-level cost. And a forward-only probe of the model’s caption-conditioned *prior* across grids gives route distances that are flat where the floors are strongly ordered, placing the ordering in J rather than in a resolution-conditioned prior or a training-distribution discontinuity.

The data branch: a route-uniform numerator over a U-shaped denominator. Measuring the mean prediction residual $\bar{r} = \mathbb{E}_\epsilon[\hat{v} - (\epsilon - x)]$ per grid and σ (the exact r in $g = \bar{J}^T r$; by Eq. 6 intrinsic posterior bias plus the frozen model’s approximation error, not model error alone) shows a strong, monotone σ -shape (cross-grid excess $0.89 \rightarrow 0.36$ in relative L_2 from $\sigma=0.125$ to 1) that is the *same for every route* within ± 0.02 , across routes whose gradient floors span 0 to 0.3 (Appendix Table 12). This kills the residual-level prediction of §3.2: with no cross-frequency coupling, that mismatch is confined to the destroyed band and ordered by its energy, which differs substantially across these routes. Route-uniformity says the real mismatch is carried by cross-band structure the diagonal model cannot express, and that all route identity lives in J — a pre-registered contrast that would equally have falsified our own assumption (iv) had the residual curves been route-ordered;

instead it *establishes* the shape factorization directly. The decay is smooth, Wiener-like shrinkage with no hard gate at the spectral crossover.

The denominator then does real work: the total gradient norm is U-shaped in σ ($2.6 \rightarrow 0.4$ at $\sigma \approx 0.3 \rightarrow 9.3$; Fig. 2b) — large at high σ where the model pulls composition out of the prior, small in the mid- σ refinement regime, inflated again at low σ by irreducible ϵ . A monotone absolute mismatch over a U-shaped norm yields exactly the measured interior-peaked curves: bin-mean gap tracks $1/\|\tilde{g}_{\text{src}}\|$ (Spearman +0.55 to +0.90 across arms, runs, and within images), and renormalizing by $\|\tilde{g}_{\text{src}}\|$ flips the anomalous low- σ dip into the monotone maximum in every arm of both runs. This is a post-hoc consistency analysis (so labeled): it explains the *shape*; all safety criteria stay in cosine units, since direction is what a normalized optimizer consumes.

The governors: ratio sets the amplitude, absolute size sets the floor. With the floor isolated, what orders the routes? Two probes give a clean division of labor, and together they decide the equal-ratio disagreement of §3.3. *Absolute size, not ratio, sets the floor*: re-run one tier down, the route 896→768 fails its pre-registered bar with a high- σ residual $\sim 2\times$ the 1024→896 plateau, at ratio 0.857 against the passing 0.875 — two boundaries the spectral family puts within 0.01 of each other at every tolerance (Appendix Table 11) — while the two routes sharing the $\sim 2,160$ -token target grid, 1024→768 (ratio 0.75) and 896→768 (ratio 0.857), land on the same floor (0.06–0.09 vs 0.06–0.12). Near-equal ratio, different target: different verdict; same target, different ratio: same floor — refuting any pure ratio rule for the floor, and with it the spectral account’s only degree of freedom. And the floor grows monotonically as the *target* grid shrinks (0.02–0.04 / 0.06–0.09 / ≈ 0.30 at target $\sim 3,000/2,160/1,000$ tokens), consistent with “floor = how well the coarse graph approximates the fine graph’s computation.” *Ratio, not absolute size, sets the amplitude*: a synthetic iso-severity route 1280→1120 (edge ratio 0.875 exactly matched to 1024→896, but $1.6\times$ its absolute target size) reproduces the 1024→896 curve bin-for-bin and its crossover $\sigma^* \approx 0.5$ (Appendix Table 10). The pre-registered contrast discriminated cleanly. Note what survives of the spectral account here: ratio is the right governor for the data term; its error is not its governor but its scope, mistaking one term for the whole gap.

The floor’s own parts: $\text{Floor}_e = \text{RoPE}_e + \text{Resid}_e$. The prediction of §3.3 — that the positional share is a coordinate-system artifact — is testable by an origin-side intervention: evaluate the demoted grid at positionally-interpolated fractional rotary coordinates (patch i at $i \cdot H_{\text{src}}/H_{\text{dem}}$), making its relative phase geometry exactly match native. At the $\sigma=1$ endpoint this *erases* the 768 floor (paired $\Delta = +0.081$, 78% of images improved) and removes $\sim 30\%$ of the 512 floor ($\Delta = +0.096$), leaving the in-band 896 control untouched: re-anchored against the debiased floors, $\text{Resid}_{768} \approx 0$ while 512 retains a $\text{Resid}_{512} \approx 0.2$ bulk. The mild-route floor is mostly a coordinate-system artifact; the harsh-route bulk is a genuine non-positional graph residue, and the capacity governor attaches to it, as the band-counting sketch anticipated. The erasure is *not* a training lever — with image content in the input the stretched forward is off-manifold — but a frequency-*selective* banded alignment survives in-window (§5.2). Full numbers, the per-block/per-module depth localization behind this design, and the inference-side prior art on the attention-over- N piece (Li et al., 2025; Liu et al., 2026) are in Appendix E.

5 TRAINING ON DOWNSCALED IMAGES

5.1 THE MEASURED (ROUTE, σ) MAP

The answer to the title question at our operating point is not a corrected formula but a measured map at a stated resolution: the curves both accounts were scored on carry the verdict, read per bin against the safety statement of Eq. 11. Figure 2 gives the debiased map for 1024-tier natives demoted to 896/768/512 (per-bin numbers: Appendix Table 6). Three reads, against criteria frozen before the runs. (i) σ -dependence is real and in the predicted direction, and tier ordering holds at every bin: the *qualitative* spectral intuition survives. (ii) The crossover for the mildest route is $\sigma \approx 0.5$ — the 896 route is clearly gapped through $\sigma \leq 0.31$, marginal at 0.44, and its means sit within ± 0.03 of zero from 0.56 on — $3.5\times$ the equal-power prediction. (iii) The $2\times$ downscale route is *certifiably unsafe* across the noise axis: its paired debiased lower bound stays above the margin in eight of nine bins (means +0.08 to +0.60), including $+0.17 \pm 0.06$ at $\sigma = 0.94$, where the input is $\sim 97\%$ noise by power

in every band the coarse grid cannot represent; the one exception ($\sigma = 0.81$, whose interval widens to include zero) certifies nothing either way — its upper bound is 0.17. There is no noise level at which a 2 $\times$ -downscaled latent certifies, or comes near certifying, the native gradient direction.

Debiasing moved verdicts in both directions, and two revisions against the historical record are part of the result. The 768 route’s high- σ plateau was roughly half estimator bias: debiased means sit within noise of zero across $\sigma \in [0.56, 0.94]$, but with upper bounds that do not cross below the $\varepsilon = 0.02$ margin at this (N, D) , and a real endpoint floor of $+0.092 \pm 0.012$ remains, so the earlier verdict of “unsafe at every noise level” softens to *no window certifiable at the practical margin* — the margins its window bins do certify (≈ 0.09 simultaneous) would not even separate the route from its own measured endpoint floor. Conversely the 896 endpoint shows a small *real* gap ($+0.042 \pm 0.011$) that single-estimate variance had buried; its safe window is the interior $\sigma \in (0.5, 0.94]$, not the endpoint (Appendix Table 9).

Collected route by route — *safe* meaning Eq. 11 holds at the stated margin, on the *per-example* object — the map needs no table: 1024 $\rightarrow$ 896 (ratio 0.875, floor 0.02–0.04) is **safe** on the interior window $\sigma \in (0.5, 0.94]$, certified at margin 0.07 (simultaneous over the four window bins; 0.056 pointwise); 1280 $\rightarrow$ 1024 (ratio 0.80, floor ≈ 0) is **safe** from $\sigma \gtrsim 0.75$, and the synthetic iso-severity grid 1280 $\rightarrow$ 1120 is safe with $\sigma^* \approx 0.5$, as its ratio twin predicts (both 1280-tier reads on the pre-debiasing instrument; §6); 1024 $\rightarrow$ 768 and 896 $\rightarrow$ 768 (floors 0.06–0.09 and 0.06–0.12) have *no certifiable window*; and any route to 512 (floor ≈ 0.30) is *certifiably unsafe* across the axis. Safety is a per-route boundary, jointly determined by ratio (through a_e) and absolute target size (through Floor_e); neither alone predicts the ordering. For the corpus safe route the fine print is: every window bin’s debiased mean sits within ± 0.03 of zero, the pointwise and simultaneous one-sided bounds certify 0.056 and 0.07 respectively, and the strict $\varepsilon = 0.02$ margin is *not yet powered* on the interior grid ($\varepsilon^* = 0.03$ –0.08 per bin). One bin’s upper bound does cross it ($\sigma = 0.69$, $\text{UB}_{95} = 0.006$, on a negative point estimate), but by the protocol of §4.1 an underpowered crossing is reported, not counted; powering the interior window at the strict margin is a draw-count exercise, designated for the next probe campaign.

A corollary worth stating plainly. The safety of 1024 $\rightarrow$ 896 is an *empirical smoothness statement about this network’s function across nearby token counts*, not an information-theoretic statement about the data. There was no a priori reason to expect a universal ratio invariant, and the spectral account’s deepest error is having promised one.

The map depends on the aggregation object. Per §3.1 the aggregation operator is part of the estimand, and the two objects genuinely disagree at high σ : pooled-arm probes (pool size 4) collapse the 1024 $\rightarrow$ 896 excess to ≈ 0 at every bin $\sigma \geq 0.625$ and the 1024 $\rightarrow$ 768 excess to ≈ 0 at $\sigma \geq 0.875$ — the batch-aggregate object SGD follows is more forgiving here than the per-example read. We publish the per-example map as the verdict (pooled arms do not yet have self-floors): the pooled collapse says the $1/B$ term dominates at these bins, but the intercept — the coherent drift no batch size removes — is bounded only by an $a + b/B$ fit across pool sizes, the designated readout of a debiased verdict grid at the shipped trainer’s actual batch $\times$ accumulation operating point [**pending**].

5.2 σ -CONDITIONAL TRAINING ON THE MEASURED-SAFE ROUTE

We wire the one measured-safe corpus route into the LoRA trainer as an opt-in path, exactly as measured, with no extrapolation beyond the map. The intervention is deliberately small: three lines inside an otherwise standard flow-matching step, and no change to the objective, the optimizer, or the noise density (pseudocode in Table 2).

σ -first draw. The trainer draws each batch’s σ *before* fetching latents, from the unchanged training density (demotion conditions on σ , not the reverse). When every sample in the batch is above the gate ($\sigma > 0.5$), the native 1024-tier latent is swapped for a cached demoted-grid sibling produced by the probe’s own measured-safe recipe (pixel-space downscale, VAE re-encode); everything downstream derives from the swapped latent. Off-route images always train native, as does validation. One honest asymmetry between the gate and the map must be stated: the gate is one-sided ($\sigma > 0.5$) while certification stops at $\sigma = 0.94$, and the endpoint carries a small real gap ($+0.042 \pm 0.011$). Since the expected gradient should vary continuously in σ , that gap cannot be dismissed as a measure-zero boundary point — it implies a neighborhood, of unmeasured width, over which some of it

Table 1: Sample-level footprint vs. the seed lottery: paired PE-Core cosine at matched (prompt, noise seed) against the *native* twin of the same training seed, mean $\pm$ seed spread (SD over the three matched-seed pairs; seed pairs for the yardstick row) over the frozen prompt set, on two single-artist corpora (60/15 training images). The yardstick row is the same cosine between native checkpoints that differ *only* in training seed; an arm reading at or above it perturbs endpoint samples less than a seed change does. **Bold:** the closest-to-native arm per corpus; the shipped path is the only arm at or inside the yardstick on both corpora.

arm (vs. native, matched seed)	corpus A	corpus B
native, cross-seed (yardstick)	0.9558 $\pm$.0046	0.9541 $\pm$.0093
896, $\sigma > 0.5$, aligned	0.9551 $\pm$.0057	0.9641 $\pm$.0023
896, every σ	0.9504 $\pm$.0179	0.9500 $\pm$.0141
768, $\sigma > 0.5$, aligned	0.9503 $\pm$.0013	0.9553 $\pm$.0127
768, every σ	0.9527 $\pm$.0118	0.9545 $\pm$.0127

Table 2: The shipped step, in full. Lines 3–5 (colored) are the entire change: the gate $\sigma^*=0.5$ and the route are read off the measured map (§5.1), line 5 is the optional refinement, and lines 1–2 are only a re-ordering of a standard step — σ is drawn before the latent is fetched.

one LoRA training step

```

1  $\sigma \sim p_{\text{train}}$ 
2  $x \leftarrow$  native latent of  $y$ 
3 if  $\sigma > \sigma^*$  and  $y$  on-route:
4    $x \leftarrow$  demoted latent of  $y$ 
5   rotary  $\leftarrow$  banded( $\sigma$ )
6  $\epsilon \sim \mathcal{N}(0, I)$  at  $\text{shape}(x)$ 
7  $z \leftarrow (1-\sigma)x + \sigma\epsilon$ 
8  $v \leftarrow \epsilon - x$ 
9 step on  $\|\hat{v}_\theta(z, \sigma, c) - v\|^2$ 

```

survives, and the tail $\sigma \in (0.94, 1)$ is exactly the interval the verdict grid does not resolve. Two facts bound the exposure meanwhile: the last interior bin reads $+0.004 \pm 0.021$, and the exercise grid of Table 1 trained *with* the tail demoted and still landed inside the seed lottery, so any tail effect is below that instrument’s resolution at 480 steps. A dense upper-tail sweep ($\sigma \in (0.9, 1)$ in four bins plus the endpoint, at higher draw count) is designated to resolve the shape; until it lands, the certified condition is $\sigma \in (0.5, 0.94]$, and the unbounded gate should be read as the form we exercised, not the form the map certifies.

Banded rotary alignment, σ -gated. Motivated by §4.2, a frequency-selective alignment (YaRN-style bands: full positional-interpolation stretch for rotary bands with few rotations across the demoted extent, native spacing for high-frequency bands, ramp between, thresholds scheduled by a sigmoid gate in σ following Zhao et al., 2026) is applied on demoted forwards only. Unlike uniform interpolation it is *not* off-manifold in-window: it passed both pre-registered legs, erasing the static variant’s low- σ liability ($+0.064 \rightarrow +0.033 \pm 0.025$) while preserving its high- σ gains (paired -0.05 vs plain demotion at $\sigma \geq 0.59$). It is a refinement candidate, not a gate-widener: the gate stays $\sigma > 0.5$. Under bit-exact deterministic training it adds no weight-space displacement over the demotion arm it refines (base $\leftrightarrow$ refined 0.319 vs base $\leftrightarrow$ demoted 0.320; without determinism a twin-run chaos floor of 0.41 would swamp this read).

Cost accounting. With $\sim 50\%$ of draws above the gate and the demoted forward at $\sim 0.72\times$ tokens, the fixed-step epoch cost is ≈ 0.86 , a *projected ceiling* of $\sim 14\%$ wall-clock on our corpus (96% of records are on the safe route), scaling with high-tier content. On the fixed-step exercise grid below the realized saving is -15.1% forward FLOPs (exact operator-level counts over each run’s realized token histogram) and -14.6% wall-clock ($n=3$ seed means on two single-artist corpora; wall tracks FLOPs $\approx 1:1$): the projection is achieved. The claim is wall-clock at fixed steps, not “more steps in the same time.”

Sample-level footprint. A fixed-step exercise grid (5 arms $\times$ 3 training seeds $\times$ 2 single-artist corpora, 480 optimizer steps each, bit-exact deterministic paired-RNG so every arm sees identical draws) renders a frozen prompt set at matched (prompt, noise-seed) pairs per checkpoint and embeds each render with a frozen vision encoder (PE-Core; Bolya et al., 2025). Table 1 reads each arm’s paired cosine against its native twin, judged against a seed-noise yardstick. The shipped path is the only arm at or inside the yardstick on both corpora (0.9551 vs 0.9558, a tie at instrument resolution; 0.9641 vs 0.9541): swapping every $\sigma > 0.5$ step to the demoted sibling perturbs endpoint samples about as much as changing the training seed. All three controls fall below the yardstick on at least one corpus, an ordering consistent with the map; but the margins are small relative to the seed spreads, so this instrument certifies the absence of a gross sample-level footprint, not quality non-inferiority.

What this section does not claim. The generation-quality acceptance gate (CMMD, Jayasumana et al., 2024, non-inferiority at training scale against the realized wall-clock saving) remains open: the

exercise grid’s first scoring pass used rating-restricted display prompts against mismatched reference pools and could not separate even the off-map control from native, so we read no quality verdict from it, and the pre-committed full-band rescoring closes the line (a regression closes it negatively). Throughput and sample-level footprint above are measured; quality non-inferiority is not. We flag this deliberately: the gap between a 480-step exercise grid and production-scale training is exactly the kind of gap this paper is about. For the same reason we deploy the *measured* map, not the predicted one — a full-pipeline bootstrap of Appendix F exposes the prediction pipeline’s own branch fragility (Appendix E).

6 LIMITATIONS

One model family, one operating point. All gradient probes ran on one DiT (Anima) with one adapter checkpoint trained at native tiers. A mixed-resolution-trained adapter might equalize (or widen) its own gradients; probing such a checkpoint is the designated bound on how far the map generalizes, as is re-probing 1280→1024 after fine-tuning at the 1280 tier. The probe pool’s relationship to the adapter’s own fine-tune set was measured rather than conceded: a controlled 2×2 factorial (two LoRAs trained on opposite style clusters under frozen membership manifests, probed on both) replicates the map’s shape on both checkpoints with the adapter×probe-style interaction below instrument resolution, while the absolute redraw-floor level is checkpoint-dependent (Appendix H).

Gradient-level, not end-task. The map is built from accumulated per-step gradients; integration over an optimizer trajectory is addressed by the exercise grid’s weight-space and sample-level reads at 480 steps rather than production scale, and the generation-quality (CMMD) gate remains open.

Account resolution and domain. The account predicts held-out routes at ~0.07–0.09 RMSE, not within ε^* ; its $A(\text{ratio})$ governor rests on two distinct ratio values; the reduction’s domain excludes the 768 mid- σ window, whose mechanism the vector-resolved probe has measured (negative interference) but which the two-term form still does not predict; the floor law’s functional form is unidentified at our operating points; and the band-counting sketch for $\text{RoPE}_e/\text{Resid}_e$ is calibrated, not derived. The graph-floor reading of the endpoint is also estimand-specific: the target-mediated perturbation is real at the vector level and lands parallel to the native gradient (Appendix E), so a magnitude-sensitive estimand — an un-normalized optimizer, say — could apportion the endpoint gap differently than the angular read does.

Instrument resolution and coverage. “Safe” is one-sided non-inferiority against a fixed margin ($\varepsilon = 0.02$ strict; the interior window currently certifies at 0.056–0.07), with the certification resolution ($\varepsilon^* = 0.03$ –0.08 per bin; ≈ 0.01 –0.02 at the endpoint sweeps) setting the power available, not the margin. True gaps below ε^* are undetectable by construction, σ^* boundaries are localized only to a bin, and every “no certifiable window” claim is (N, D) -relative: a larger campaign could certify (or refute) the 768 high- σ window at the strict margin. The upper tail $\sigma \in (0.94, 1)$ of the safe route is unmeasured between the last interior bin and the endpoint’s real gap; the designated dense sweep (§5.2) closes it. Verdict stability under the per-image trim rule of §4.1 has not been separately audited. The debiasing campaign covered the corpus verdict grid; the 1280-tier iso-severity, 896-tier transfer, depth-split, and PI-intervention probes remain pre-debiasing reads, and the 1280-tier curves enter the held-out validation on that basis.

Scope. Adapter (LoRA) gradients on a frozen backbone; full fine-tuning may differ. Pixel-space demotion only (latent-space downsampling was not tested here; SwD found it inferior). The σ -gated banded-alignment parameters were taken from one shot at pre-registered values, not searched.

7 CONCLUSION

When does training on downsampled images yield the same gradient direction? Not when the noise has masked the frequencies the coarse grid loses. That spectral answer was advanced by prior work for inference-time substitution and had not previously been confronted with the training gradient; transported to the training question it is a model of only one part of one branch of what demotion perturbs, and the measurements reject it at every point of structural divergence — the mildest route’s crossover sits at 3.5× the equal-power prediction, no tolerance reproduces the map, its one free parameter is inert at the curve level, and a 2×-downsampled latent certifies at no noise level, because

the network function itself is grid-calibrated: partly through its positional coordinate system, partly through an irreducible dependence of the computation on token count.

The measured answer has two conditions, matching the two terms of our account: the target grid must be large enough in *absolute token count* that its graph floor sits below certification resolution, and the noise level must exceed a per-route boundary where the ratio-governed data term dies. On our model that means $1024 \rightarrow 896$ on $\sigma \in (0.5, 0.94]$ (the upper tail toward the endpoint’s small real gap still unresolved) and $1280 \rightarrow 1024$ at $\sigma \gtrsim 0.75$, and a $2\times$ downscale at no noise level. The answer is predictive, not merely descriptive: head to head on the same curves the account predicts held-out routes at $\sim 0.07\text{--}0.09$ RMSE against the spectral family’s $0.147\text{--}0.355$, beating an oracle fit on the held-out data itself on one route. Getting there required treating the question as a metrology problem: naive finite-draw estimation manufactures exactly the token-count-ordered floor signature under test, half the intermediate route’s raw floor was estimator bias, and our own headline claims were re-derived, and in one case softened, under the correction. On the one route our corpus makes safe, the trainer realizes the map’s projected ceiling (-14.6% wall-clock at fixed steps) with a sample-level footprint within the seed lottery; quality non-inferiority at production scale remains the open gate. We offer the account and its instrument as the cheap, general test that any future “train part of the time at lower resolution” proposal should pass before it is believed — and without the debiasing, that test manufactures the very effect it looks for.

REPRODUCIBILITY STATEMENT

The probe scripts (`run_sigma_probe.py`, including the `--self_floor` and `--draw_sweep` debiasing modes; `run_prior_distance.py`, `rapd.py`, `compare_ckpt_dw.py`), the held-out validation and refit analyses (`e5_holdout.py`, `e5_refit.py`, `e83_bridge.py`), the trainer wiring (`--sigma_lowres`), and the frozen pre-registration documents are in the public repository at https://github.com/sorryhyun/anima_lora (project line `project/sigma_lowres/`), together with the debiased verdict runs’ complete per-image records. The earlier raw-instrument archives are excluded by the repository’s results ignore rule and will be published as a companion tarball [**pending**]; until then those numbers are reproducible from the scripts but their original records are not public. Paired training runs use a `--deterministic` mode verified bit-identical across twin runs; finite-draw cosines are kernel-path-sensitive, so all cosine pairings are computed within a single run (Appendix E).

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A NOTATION AND ESTIMAND FINE PRINT

Arm labels versus route subscripts. One notational convention holds throughout. Arm labels (src, reenc, dem) mark *whose* gradient, latent, or Jacobian an object is; the subscript on every route-level scalar is the *route*, written in full ($d_{e_0 \rightarrow e}$) where several sources appear, and abbreviated to the target edge alone — an edge value, never an arm label, so it always instantiates numerically (d_e , Floor₁₁₂₀, Resid₇₆₈) — wherever the source edge is fixed by context, as it is in almost every measured statement. The control’s distance d_{reenc} (Eq. 1 with $\bar{g}_{\text{reenc}}$ in place of $\bar{g}_{\text{dem}}$) is the one exception that carries an arm label: the control changes no grid and has no target edge. Each table states which object it reports, d_e or the excess gap _{e} .

B DERIVATION OF THE BRANCH DECOMPOSITION (EQ. 5)

Fix a query (image, caption), a σ , and an arm a . Per draw ϵ , the adapter gradient of the $\frac{1}{2}$ -scaled loss is the product $g_a(\epsilon) = J_a(\epsilon)^\top r_a(\epsilon)$ of §3.3. Split each factor into its noise mean and fluctuation, $J_a = \bar{J}_a + \tilde{J}_a$ and $r_a = \bar{r}_a + \tilde{r}_a$ with $\mathbb{E}_\epsilon[\tilde{J}_a] = 0$ and $\mathbb{E}_\epsilon[\tilde{r}_a] = 0$; the population mean gradient of arm a is then

$$\bar{g}_a = \mathbb{E}_\epsilon[J_a^\top r_a] = \bar{J}_a^\top \bar{r}_a + \mathbb{E}_\epsilon[\tilde{J}_a^\top \tilde{r}_a]. \quad (12)$$

Write the demoted arm’s mean factors as native plus a perturbation, $\bar{J}_{\text{dem}} = \bar{J} + \Delta J$ and $\bar{r}_{\text{dem}} = \bar{r} + \Delta \bar{r}$ (bars without an arm subscript denote the native arm). Subtracting Eq. 12 across arms and expanding the product,

$$\delta g = \bar{g}_{\text{dem}} - \bar{g}_{\text{src}} = \underbrace{\bar{J}^\top \Delta \bar{r}}_{\text{data branch } B_e} + \underbrace{\Delta J^\top \bar{r}}_{\text{graph branch } C_e} + \underbrace{\Delta J^\top \Delta \bar{r} + \mathbb{E}_\epsilon[\tilde{J}_{\text{dem}}^\top \tilde{r}_{\text{dem}}] - \mathbb{E}_\epsilon[\tilde{J}_{\text{src}}^\top \tilde{r}_{\text{src}}]}_{\text{remainder } R_e}. \quad (13)$$

This is an exact identity: no linearization is involved. “First-order” in the main text refers only to the content of R_e : it collects every term that is a product of two perturbations ($\Delta J^\top \Delta \bar{r}$) or a difference of noise-covariance terms (the last two). Eq. 5 is Eq. 13 with the mean-Jacobian bar suppressed in the notation.

One remark delimits the identity’s literal scope. When the two arms share a grid (as for the re-encoding control), every operation above is literal. Across grids, J_{src} and J_{dem} act on different token counts, so the differences ΔJ and $\Delta \bar{r}$ require a fixed identification of the two output spaces (e.g. spectral zero-padding of the coarse grid onto the fine grid’s bands); any such identification changes the split between B_e , C_e , and R_e but not their sum δg , which lives in the shared adapter-parameter space regardless. This is why the main text defines the branches *operationally* (as the two independent perturbation directions in adapter-parameter space, isolated by the designated probes of §4.2) and treats Eq. 5 as fixing what each branch collects rather than as a computable matrix formula.

C THE MEAN RESIDUAL AS A POSTERIOR OPERATOR (EQ. 6)

This appendix derives the posterior form of the fixed-image mean residual, the exact split of the measured object into intrinsic posterior bias plus model approximation error, and the Gaussian closure referenced in §3.3. Fix a caption c and grid e , write $a = 1 - \sigma$, and let $x \sim p_e(\cdot | c)$ be the grid- e clean latent, $z_{\sigma} = ax + \sigma\epsilon$ with $\epsilon \sim \mathcal{N}(0, I)$.

Bayes field and fixed-image residual. For the squared objective the population-optimal field is the conditional mean of the target $v = \epsilon - x = (z_{\sigma} - x)/\sigma$:

$$v_e^*(z, \sigma, c) = \mathbb{E}[v | z_{\sigma} = z, c] = \frac{z - m_e(z, c)}{\sigma}, \quad m_e(z, c) := \mathbb{E}[x | z_{\sigma} = z, c]. \quad (14)$$

Averaging over noise at fixed image ($\mathbb{E}_\epsilon[\epsilon] = 0$, $\mathbb{E}_\epsilon[z_{\sigma}] = ax$) gives Eq. 6:

$$\bar{r}_e^*(\sigma; x) = \mathbb{E}_\epsilon[v_e^*(ax + \sigma\epsilon, \sigma, c) - (\epsilon - x)] = \frac{x - \mathbb{E}_{z_{\sigma}|x}[m_e(z_{\sigma}, c)]}{\sigma}. \quad (15)$$

With $p_{\sigma,e}(z | c)$ the noised marginal, Tweedie’s identity for the linear interpolant reads $m_e(z, c) = (z + \sigma^2 \nabla_z \log p_{\sigma,e}(z | c))/a$ for $\sigma < 1$, so equivalently

$$\bar{r}_e^*(\sigma; x) = -\frac{\sigma}{1 - \sigma} \mathbb{E}_{z_{\sigma}|x}[\nabla_z \log p_{\sigma,e}(z_{\sigma} | c)], \quad (16)$$

which makes the missing ingredient explicit: the numerical curve requires the grid- and caption-conditional smoothed score, i.e. a model of $p_e(x | c)$. At $\sigma = 1$, where $z = \epsilon$ is independent of x , $m_e(z, c) \rightarrow \mathbb{E}[x | c]$ and $\bar{r}_e^*(1; x) = x - \mathbb{E}[x | c]$. With A_e the alignment operator that places the source-grid residual on the demoted grid (§4.1), the Bayes-level cross-grid mismatch is $\Delta \bar{r}_e^*(\sigma; x) = \bar{r}_{\text{dem}}^*(\sigma; x_{\text{dem}}) - A_e \bar{r}_{\text{src}}^*(\sigma; x_{\text{src}})$.

What the instrument measures. MSE optimality gives $\mathbb{E}[v^* - v | z, c] = 0$ and hence a zero residual after a *joint* average over (x, ϵ) ; it does not zero the fixed-image average $\mathbb{E}_\epsilon[v^* - v | x, c]$, which is Eq. 15 and generally nonzero. Writing the trained network as $\hat{v}_{\theta, e} = v_e^* + q_{\theta, e}$, the probe’s per-image object splits exactly as

$$\bar{r}_{\theta, e}(\sigma; x) = \bar{r}_e^*(\sigma; x) + \mathbb{E}_{z_\sigma | x}[q_{\theta, e}(z_\sigma, \sigma, c)] : \quad (17)$$

intrinsic posterior reconstruction bias, derivable in principle from $p_e(x | c)$, plus genuinely model-specific approximation error. The measured $\|\Delta \bar{r}(\sigma)\|$ therefore has definite theoretical status — an empirical closure of Eq. 15 — but is not a pure model-error difference, and no claim in the paper requires it to be one: the two-term account consumes the measured curve as-is (assumption (iv)), and the trained-model residual is in any case the exact r -factor that the LoRA gradient $g = J^\top r$ is built from.

Gaussian closure. The cheapest nontrivial closure is a full Gaussian model $x | c \sim \mathcal{N}(\mu_e, C_e)$ per grid. With $Q_e(\sigma) = a^2 C_e + \sigma^2 I$, the posterior mean is linear, $m_e(z, c) = \mu_e + a C_e Q_e^{-1}(z - a \mu_e)$, and substituting into Eq. 15 collapses (via $I - a^2 C_e Q_e^{-1} = \sigma^2 Q_e^{-1}$) to

$$\bar{r}_e^*(\sigma; x) = \sigma Q_e(\sigma)^{-1}(x - \mu_e). \quad (18)$$

For paired source and demoted latents with covariance blocks C_{ss}, C_{dd}, C_{ds} and $L_s = \sigma Q_s^{-1}$, $L_d = \sigma Q_d^{-1}$, the image-population mean squared mismatch is

$$\mathbb{E}\|\Delta \bar{r}_e^*\|^2 = \text{tr}(L_d C_{dd} L_d^\top) + \text{tr}(A_e L_s C_{ss} L_s^\top A_e^\top) - 2 \text{tr}(L_d C_{ds} L_s^\top A_e^\top), \quad (19)$$

a σ -curve computable from paired *clean latents only*, with the source–demote cross-covariance C_{ds} load-bearing. A *diagonal Fourier* C_e makes Eq. 18 the per-band Wiener shrinkage of §3.2, confining the mismatch to the destroyed band and ordering it by destroyed-band energy — the residual-level prediction that route-uniformity falsifies (§4.2). We tested whether any structured second-order closure suffices, entirely on stored paired clean latents (no denoiser forward): three nested covariance models (diagonal spectral; + within-band channel blocks; + octave-pair cross-band coupling), fitted on held-out-from-probe images and scored on the probe set against the measured curves. All three reproduce the curve’s monotone σ -shape (Pearson 0.94–0.97), its route-uniformity, the near-zero re-encoding control, and the $\sigma=1$ endpoint, but over-predict the low- σ amplitude by ~40%, failing a pre-registered RMSE bar — so second-order data-only structure recovers the shape but not the level, and the essential missing ingredients are the ones the identity names: caption-conditioning (the closure is caption-marginal), non-Gaussian posterior structure, and the q_θ term of Eq. 17. The paper therefore retains the measured $\|\Delta \bar{r}(\sigma)\|$ as its minimal honest closure. A theory–instrument comparison must also reproduce the instrument’s estimand — the split-half-corrected *relative* L_2 distance of §4.1, i.e. $\sqrt{\mathbb{E}\|\bar{r}_d - A_e \bar{r}_s\|^2}$ normalized by the mean of the two arms’ root-mean-square norms — not a raw mismatch norm.

D DERIVATION OF THE ANGULAR FORMS (EQS. 7 AND 8)

Both angular forms follow from Eq. 1 and the split of δg by the native direction; no property of demotion enters. Write $g = \bar{g}_{\text{src}}$, $G = \|g\|$, $\hat{g} = g/G$, and $u^\perp = u - (\hat{g}^\top u) \hat{g}$; decompose $\delta g = G \kappa_\parallel \hat{g} + \delta g^\perp$ with $\kappa_\parallel = \hat{g}^\top \delta g / G$ and $\kappa_\perp = \|\delta g^\perp\| / G$.

The exact form. The two components are orthogonal, so $\|g + \delta g\|^2 = G^2(1 + \kappa_\parallel)^2 + G^2 \kappa_\perp^2$ and

$$\cos(g, g + \delta g) = \frac{\hat{g}^\top (g + \delta g)}{\|g + \delta g\|} = \frac{1 + \kappa_\parallel}{\sqrt{(1 + \kappa_\parallel)^2 + \kappa_\perp^2}} = \frac{1}{\sqrt{1 + \kappa_{\text{eff}}^2}}, \quad \kappa_{\text{eff}} = \frac{\kappa_\perp}{1 + \kappa_\parallel}, \quad (20)$$

valid whenever $1 + \kappa_\parallel > 0$ (the perturbation does not reverse the gradient direction); $d_e = 1 - \cos$ is Eq. 7. Since $1 - \cos(\hat{g}_1, \hat{g}_2) = \frac{1}{2}\|\hat{g}_1 - \hat{g}_2\|^2$ for unit vectors, starting from either expression in Eq. 1 lands on the same result.

The quadratic limit. For small perturbations, $1 - (1 + \kappa_{\text{eff}}^2)^{-1/2} = \frac{1}{2}\kappa_{\text{eff}}^2 + O(\kappa_{\text{eff}}^4)$ and $\kappa_{\text{eff}}^2 = \kappa_{\perp}^2(1 - 2\kappa_{\parallel} + O(\kappa_{\parallel}^2))$, so

$$d_e = \frac{\|\delta g^{\perp}\|^2}{2G^2} + \underbrace{O(\kappa_{\perp}^2 \kappa_{\parallel}) + O(\kappa_{\perp}^4)}_{\text{beyond-quadratic}}. \quad (21)$$

The four-term expansion. Substituting $\delta g = B_e + C_e + R_e$ (Eq. 13) into the leading term and expanding the square,

$$\|\delta g^{\perp}\|^2 = \|B_e^{\perp}\|^2 + \|C_e^{\perp}\|^2 + 2 \langle B_e^{\perp}, C_e^{\perp} \rangle + \left(2 \langle (B_e + C_e)^{\perp}, R_e^{\perp} \rangle + \|R_e^{\perp}\|^2 \right),$$

whose first three terms, divided by $2G^2$, are exactly the data share S_e , the graph share F_e , and the projected interaction I_e of Eq. 8. The remainder R_e' of Eq. 8 therefore has an exact content: the R_e -bearing terms above, the beyond-quadratic terms of Eq. 21, and the parallel-component correction, each suppressed by an extra power of perturbation size relative to the retained terms.

From the four-term form to the excess. The reported object is the excess over the control. The reenc arm changes no grid, so $\Delta J_{\text{reenc}} = 0$, and both control terms built from it vanish identically — its graph share F_{reenc} and its projected interaction I_{reenc} each carry a factor $(\Delta J_{\text{reenc}}^{\top} \bar{r})^{\perp} = 0$. Writing Eq. 8 for each arm and subtracting term by term,

$$\begin{aligned} \text{gap}_e(\sigma) &= d_e(\sigma) - d_{\text{reenc}}(\sigma) \\ &\approx (S_e - S_{\text{reenc}}) + (F_e - F_{\text{reenc}}) + (I_e - I_{\text{reenc}}) + (R_e' - R_{\text{reenc}}') \quad \text{Eq. 8, each arm} \\ &= (S_e - S_{\text{reenc}}) + F_e + I_e + (R_e' - R_{\text{reenc}}'), \quad F_{\text{reenc}} = I_{\text{reenc}} = 0 \end{aligned}$$

so the control collapses to $d_{\text{reenc}} \approx S_{\text{reenc}} + R_{\text{reenc}}'$, with S_{reenc} the re-encode share (measured ≈ 0 where probed: the control row of Table 7). The subtraction cancels only the re-encode component of the data share (up to a cross term suppressed by $\sqrt{S_{\text{reenc}}}$, bounded by the control's near-zero reading), so $\|\Delta \bar{r}\|$ is read net of re-encode; the graph share and the interaction pass through untouched — the control has nothing to cancel them with.

The drop bounds. Both discarded terms obey exact bounds in the retained shares (Cauchy–Schwarz on the expansion above),

$$|I_e| \leq 2\sqrt{S_e F_e}, \quad |R_e'| \leq 2\sqrt{(S_e + F_e + I_e) \rho_e} + \rho_e + O(\kappa_{\perp}^2 \kappa_{\parallel}, \kappa_{\perp}^4), \quad (22)$$

with $\rho_e = \|R_e^{\perp}\|^2 / (2\|\bar{g}_{\text{src}}\|^2)$ the remainder branch's own share. The first says the interaction can never exceed the retained sum ($2\sqrt{S_e F_e} \leq S_e + F_e$) and is automatically negligible wherever one share dominates the other, so assumption (ii) carries independent content only near share crossover — which is exactly where Appendix F finds it failing. The second says the first-order part of the remainder enters at half power — one factor of $\sqrt{\rho_e}$ per retained share — so assumption (i) reduces to the branch-level statement that R_e is small against B_e and C_e , plus the small- κ domain for the tail. The control counterpart R_{reenc}' needs no bound of its own: the control's degenerate expansion is itself the measured near-zero row of Table 7, which pins R_{reenc}' jointly with S_{reenc} .

What is not derived. The loading $\|\delta g^{\perp}\| = a_e \|\Delta \bar{r}(\sigma)\|$ (that the orthogonal gradient mismatch is proportional to the residual mismatch with a σ -independent route gain) is an empirical ingredient (the data-branch factorization at the amplitude level), fit and tested held-out in Appendix F. The geometry above fixes only how a given δg is read into gap units. What *is* derivable from first principles is only a one-sided envelope, $\|\bar{J}^{\top} \Delta \bar{r}\| \leq \sigma_{\max}(\bar{J}) \|\Delta \bar{r}(\sigma)\|$, which would turn the data term into an inequality in the observed mismatch norm given a σ -uniform bound on the mean Jacobian's top singular value. Promoting the envelope to a proportionality with a σ -independent gain requires the mismatch *direction* $\widehat{\Delta \bar{r}}(\sigma)$ to hold a σ -stationary alignment with $\bar{J}$'s singular directions (isotropy, for instance, would give gain $\|\bar{J}\|_F / \sqrt{n}$) while $\bar{J}$ itself varies with σ through the network input — and the image- and route-specific mismatch directions of Appendix E license no such axiom. Hence the loading stays an assumption with a designated probe rather than a theorem.

E INSTRUMENT DETAILS AND SECONDARY READS

Estimator. Per image, arm, and σ -bin, adapter gradients are accumulated over D stratified draws and flattened; cosines are taken between accumulated vectors. Per-bin cosines at $D=8$ are not comparable *across* bins (the floor drops where $\|g\|$ is small); the gap subtraction against the same-bin floor is the valid read. Split-half reliability over images of each bin-mean curve is mandatory (observed 0.73–0.88 on verdict runs); a per-image variant of the same quantity has split-half reliability ≈ 0 and is not used.

Controls. The re-encoding arm prices the VAE decode–encode round trip that demotion necessarily pays; in pre-debiasing runs its gap served as the validity band (± 0.04 ; observed $|\cdot| \leq 0.054$ across all bins of the main runs, per-module-group ≤ 0.045 in the split runs), a role superseded by the paired debiased read of §4.1. The redraw floor prices finite-draw noise. Density-weighting the uniform bins by the trainer’s σ -density reproduces earlier σ -marginalized measurements from the same instrument family (consistency check).

Debiasing, implementation. Under `--self_floor`, every arm (re-encoding and each demote variant) runs a second independent draw set from a disjoint seed stream, giving per-arm self-cosines alongside the native floor; Eq. 10 is then computed per image and bin, and the map read is the paired excess $\text{gap}_{e,i} = \widehat{\text{gap}}_{e,i} - \widehat{\text{gap}}_{\text{reenc},i}$ with non-finite values (negative self-cosines under the square root at low D) and $|\text{gap}_{e,i}| > 1.5$ trimmed (0–5 of 40 images per cell). Under `--draw_sweep`, draw prefixes are nested ($D = 64$ contains the $D \leq 32$ sets; prefix sums, no extra forwards) and $\text{gap}(D) = \text{gap}_\infty + c/D$ is fit on route means with a bootstrap CI over images. Finite-draw cosines are kernel-path chaotic: twin runs sharing a warm compiler kernel cache agree to $|\Delta \cos| \leq 0.015$, while a run compiled to a different kernel set lands up to ≈ 0.3 away at $D=2$, so gap/floor/self-floor pairings are not compared across processes, a guarantee the single-run instrument provides by construction; a `--deterministic` mode (bit-exact across twin runs) covers the paired training A/Bs.

Finite-draw bias, measured. The bias mechanism of §4.1 is large at the probe’s operating point: a variance back-of-envelope predicts spurious iso-direction endpoint gaps of $\approx 0.02/0.05/0.15$ for 896/768/512, and the 1024→896 endpoint gap indeed decays as $+0.100/+0.035/-0.016$ at $D = 4/8/16$, a clean c/D decay with fitted $c \approx 0.5$. After debiasing, the fits come out D -flat ($|c| \leq 0.05$ for the 512 route, against $c \approx +0.29$ uncorrected). The native floor makes the point sharpest: at $D=8$ –16 the uncorrected endpoint self-cosine reads 0.79–0.85, but the nested sweep ($D = 4 \dots 64$) extrapolates it to 1.005 (bootstrap 95% CI [0.994, 1.016]) — the entire redraw floor is finite-draw noise.

Endpoint and x-zero modes. The $\sigma=1$ endpoint bin feeds input exactly ϵ (the input-mediated data term vanishes by construction). The x-zero mode zeroes x in input *and* target on every grid, keeping captions and exact demoted latent shapes; with the $(1-\sigma)x$ term absent, low- σ bins are off-manifold and the $\sigma=1$ read is primary. In that regime the residual is $\approx -\hat{x}_{\text{prior}}$, so “graph-dominated” includes the model’s grid-conditioned prior; the prior-distance probe (§4.2) subsequently dissociated the prior from the floor’s route ordering. In the target-strength (α) sweep the interior points $\alpha \in [0.5, 0.75]$ are unreadable by construction and excluded: there the native residual $\alpha x - \hat{x}$ passes near cancellation, the gradient norm dips ($60 \rightarrow 33$) and the redraw floor falls to 0.87, so the estimator reads through a small gradient norm — the low-norm amplification of §4.2 surfacing along the α axis. The 512 route’s sweep anchors are $+0.269 \pm 0.041$ at $\alpha=0$ vs $+0.337 \pm 0.067$ at $\alpha=1$, α -flat like the 768 anchors quoted in the main text.

Parallel-landing decomposition of the target term (behind §4.2). The α -flatness has an exact mechanism. Because the objective is quadratic, $\bar{g}$ is affine in α , so the exact target-mediated gradient $t = \bar{g}(1) - \bar{g}(0)$ is computable from the endpoint arms; measured, it is real and large ($\|t_{\text{src}}\|/\|\bar{g}_{\text{src}}\| \approx 2.2$, so $J^\top$ does not annihilate target content), but demotion’s change to it lands almost entirely *along* $\hat{g}_{\text{src}}$ ($\kappa_{\parallel} = -0.75/-1.18/-1.86$ against $\kappa_{\perp} = 0.09/0.14/0.20$, an 8–9× parallel dominance reproducible across draw sets). The cosine gap is blind to a parallel rescaling (Eq. 7) and the orthogonal part enters only at second order ($\kappa_{\perp}^2/2 \approx 0.004$ –0.02), so the α -flat result stands *because* the target term lands parallel, not because it vanishes. In this read the per-image orthogonal loading is $\sim 10\times$ the aggregate’s: image-specific directions cancel in the mean, the same motif as the

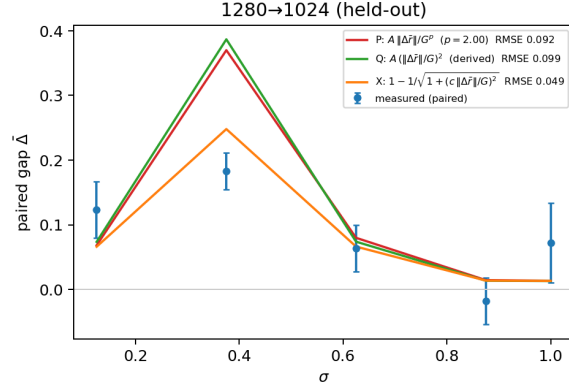


Figure 3: The second held-out route, 1280→1024, under all three functional forms for the data term of Appendix F (fitted power, derived quadratic, exact angular link; legend $G := \|\bar{g}_{\text{src}}(\sigma)\|$), predicted from governors fitted on other routes. This is the one route where the forms visibly separate: the derived quadratic systematically overshoots the mid- σ peak; the exact angular link’s saturation removes the overshoot (RMSE 0.049 vs ≈ 0.095), which is why the exact link headlines in Fig. 1.

residual-direction read below. For 896 and 768 the floors sit at or below the interior bins’ certification resolution, which the endpoint sweeps resolve only through their order-of-magnitude larger draw budget — so “the high- σ plateau is the floor” sharpens to: the floor is what the plateau converges to once the data term dies and the estimator can see it.

Direction-resolved mismatch (behind §4.2). The route-uniformity of $\|\Delta\bar{f}(\sigma)\|$ is a property of the *amplitude* only. Saving the per-image mismatch vectors and comparing $\Delta\bar{f}$ directions with a split-half attenuation correction, non-adjacent route pairs are near-orthogonal at low σ (corrected cosine 0.00–0.08 at $\sigma \leq 0.375$) with only a weak shared component at high σ (+0.2–0.3 at $\sigma \geq 0.875$); the top mode of a stacked SVD carries 0.33–0.36 of the energy against the 0.25 rank-4-uniform baseline (a rank-one common mode would give ≈ 1); and cross-image direction consistency is zero at every (route, σ) cell (max +0.019 at split-half reliabilities 0.73–0.99), so the mismatch direction is fully image-specific — refuting a grid-conditional composition prior (“small canvas $\Rightarrow$ portrait framing”) as the carrier, at least under full captions. What *is* shared is scale and spectral drift: $\Delta\bar{f}$ energy migrates toward low frequencies as σ rises (low-third share 0.40 $\rightarrow$ 0.67 by $\sigma = 0.875$), composition-scale but per-image. Eq. 9 only ever consumes the norm, so the account is untouched; what sharpens is the open question, now “why is only the *amplitude* universal when the directions are image- and route-specific.” A closed-form candidate for cross-band coupling exists — the flow posterior covariance, whose off-diagonal $D_z v^*$ entries are $-\left((1-\sigma)/\sigma^3\right) \text{Cov}(x_\omega, x_{\omega'} \mid z)$ (Xing et al., 2026), set to zero by the diagonal-Gaussian premise — but its rank-one common-mode version is excluded by the SVD read above.

Held-out validation and refit, sources. The fit consumes the debiased paired curves of the 1024-tier verdict run, the 1280-tier curves (pre-debiasing paired reads; see §6), the route-uniform mismatch norm $\|\Delta\bar{f}(\sigma)\|$ from the residual probe, and each run’s own bin-mean gradient norm $\|\bar{g}_{\text{src}}(\sigma)\|$ (gap and norm always from the same process, per the kernel-path rule). The shared-power fit scans $p \in [1, 2]$ jointly across fit routes with per-route weighted least squares; the exact-link fit is a per-route grid with profile-likelihood uncertainties on a_e . Oracle baselines are quadratic-in- σ fits on the held-out data itself (the noise ceiling for a 3-dof curve); the spectral baseline on its committed region is gap = 0 for $\sigma > \sigma_{\text{eq}}$, and its curve-level transport is the bridge of Fig. 5.

The floor law’s identifiability. Two anchors and one held-out check cannot identify a functional form, and the cosine gap is the wrong scale to fit one in: it saturates (Eq. 7). Refitting the same anchors in the unsaturated perturbation-energy units $\kappa_{\text{eff}}^2 = (1 - F)^{-2} - 1$ gives $\tau \approx 860$ tokens and predicts $F(2160) = +0.096$, equally consistent with the measured $+0.092 \pm 0.012$; the same two anchors pushed through $e^{-\sqrt{n}/\ell}$ or a power law n^{-p} predict +0.085 and +0.075. Every form lands within the union of the measured CI and the prediction’s own bootstrap spread, so the exponential is one member

of a family our operating points cannot distinguish, and the mechanical reading of τ ($\approx 860\text{--}1040$ tokens across the two unit conventions) as an effective-rank decay length of the early-block gradient operator is a hypothesis, not a law. It is a discriminable one: a spectral-tail account predicts a source-capacity dependence $F \propto e^{-n/\tau} - e^{-n_0/\tau}$ where the absolute-target-capacity governor predicts none (at our operating points the secant correction sits within calibration slack, so the existing anchors do not discriminate); the designated run is a fixed-target, varied-source token ladder, not yet scheduled.

Bootstrap bands and the leave-896-out lane (Fig. 1). The bands are a full-pipeline image bootstrap ($B=1000$, fixed seed; 981 draws kept): resample images with replacement within each run ($N=40 / N=24$), re-bin the paired curves, refit the exact-link (a_e, Floor_e) on every fit route, re-derive the ratio governor and floor law, re-predict. $\|\Delta\bar{f}(\sigma)\|$ and the bin-mean $\|\bar{g}_{\text{src}}(\sigma)\|$ are run-level instruments with no per-image decomposition and are held fixed. The replica includes the committed floor law’s positive-floor filter ($\text{Floor} > 0.005$): in 41% of draws the resampled 1120 floor clears the filter and the law re-anchors on the (896, 1120) legs rather than (512, 896) — branch variability the bands therefore contain, and the reason §5.2 deploys the measured map rather than the predicted one. The 896 curve in Fig. 1 is the post-hoc leave-896-out lane under the exact link (c from the ratio-twin 1120 alone = 0.142 vs 0.150 fitted on 896 itself; floor law through 512+1120, whose fragile Floor_{1120} leg enters the log fit clamped positive (36% of bootstrap draws clamp), giving $F(3012) = +0.012$ vs measured +0.019 [+0.010, +0.030]; curve RMSE 0.073).

Interventional B/C ledger, implementation. Routes $1024 \rightarrow \{896, 768, 512\}$, $\sigma \in [0.5, 1.0]$ in four bins plus the $\sigma=1$ endpoint bin, $D=8$ draws per bin, $N=24$ images, deterministic mode, two independent draw sets per arm. Each arm’s draw-summed adapter gradient is stored per bin; the interventional split is $B = \bar{g}_{\text{repmote}} - \bar{g}_{\text{src}}$ (the data intervention at fixed native graph: the demote-re-promote arm’s input passed through the downscale $\rightarrow$ upscale $\rightarrow$ encode pipeline but evaluated on the native grid) and $C = \bar{g}_{\text{demote}} - \bar{g}_{\text{repmote}}$ (the graph intervention at fixed demoted data). The quadratic shares S, F, I and $\rho = \cos(B^\perp, C^\perp)$ use cross-draw-set inner products to cancel shared draw noise; the debiased estimates are not confined to $[-1, 1]$ (hence $\rho = -1.24$ at one low-amplitude bin). The exact counterfactual angles $h(B), h(C), h(B+C)$, with $h(X) = 1 - \cos(\bar{g}_{\text{src}}, \bar{g}_{\text{src}} + X)$, are computed on the raw arm means; since $\bar{g}_{\text{dem}} = \bar{g}_{\text{src}} + B + C$ by construction, the realized demotion distance is exactly $h(B+C)$, so each account’s predicted-to-realized ratio is direct. The no-interference scalar account $S+F$ overpredicts the realized in-window gap $2.4\text{--}3.8\times$ at 768 ($1.6\text{--}5.8\times$ at 896, $1.2\text{--}3.0\times$ at 512); the fully additive counterfactual $h(B) + h(C)$ overpredicts $3.5\text{--}8\times$, the realized $h(B+C)$ being only $\sim 20\text{--}30\%$ of the additive sum in-window. Including I is what removes the overprediction; the vector account $B+C$ is exact by construction. One unit-honesty note on Table 3: in-window $|B^\perp|, |C^\perp| \approx 0.5\text{--}1.0 \|\bar{g}_{\text{src}}\|$, outside the quadratic truncation’s domain, so the truncated ledger $S+F+I$ underpredicts realized magnitudes $\sim 3\text{--}4\times$ (median ratio $0.24\times$). The ledger licenses sign, decomposition, and window localization; gap magnitudes are quoted from $h(\cdot)$. Finally, the re-encoding proxy check re-references the data leg against the control arm, $B_{\text{reenc}} = \bar{g}_{\text{repmote}} - \bar{g}_{\text{reenc}}$: $|B_{\text{reenc}}^\perp|$ agrees with $|B^\perp|$ to $\sim 4\%$ at the signal-carrying low- σ bins (worst $\pm 23\%$ only where B is already small), so the shared downscale $\rightarrow$ encode pipeline cost is a minor share of the data intervention — closing the one open item on the floor ledger’s first share, where the re-encoding control (native decode $\rightarrow$ re-encode) is a *proxy* for the pipeline cost demotion actually pays.

Floor decomposition, detail (behind §4.2). Splitting the probe gradients per module type (15 groups, including separate rows of the fused QKV projections) and per block (28) localizes the floor in *depth*, not module type: early blocks (0–9, peaking at 3–8) carry $\sim 3\times$ the late-block gap, uniformly across every module type within a block (at $512/\sigma=1$ every type sits in $0.22\text{--}0.31$; Table 13). The mechanism picture: the first ~ 10 blocks build grid-calibrated token statistics, the divergence propagates into every parameter type’s gradient in those blocks roughly equally, and deeper blocks inherit a washed-out version. Query and key projections show *zero* excess over value projections, which we initially read as refuting a rotary mechanism — an instructive error, since landing-side uniformity cannot localize *origin*: a perturbation originating in the position embedding propagates through the block and lands on all module types. Hence the origin-side positional-interpolation intervention (Table 14; the paired Δ s are same-grid contrasts at matched draws, in which the finite-draw bias cancels to first order). The 896 route’s small floor sits below that probe’s paired resolution and remains undecomposed. A σ -resolved follow-up closes the tempting route this opens (training $1024 \rightarrow 768$ through interpolated coordinates): with image content in the input the stretched forward is off-manifold, and

Table 3: The interventional B/C ledger behind Appendix F: per (route, σ -bin) orthogonal amplitudes (units of $\|\hat{g}_{\text{src}}\|$), interaction geometry, the cross-set-debiased quadratic shares, and the exact counterfactual angles. In-window $S+F+I$ is sign- and localization-accurate but underpredicts $h(B+C)$ (small-perturbation truncation out of domain); magnitude claims in the text use $h(\cdot)$.

route	σ	$ B^\perp $	$ C^\perp $	ρ	S	F	I	$h(B)$	$h(C)$	$h(B+C)$
896	0.563	0.55	0.59	-0.95	0.103	0.115	-0.206	0.107	0.206	0.037
	0.688	0.42	0.39	-0.91	0.036	0.059	-0.085	0.083	0.077	0.062
	0.813	0.20	0.23	-0.94	0.016	0.019	-0.033	0.022	0.027	0.010
	0.938	0.10	0.10	-1.24	0.002	0.003	-0.006	0.007	0.006	0.003
	1.000	0.05	0.06	-0.71	0.001	0.001	-0.001	0.002	0.003	0.002
768	0.563	0.57	0.68	-0.93	0.139	0.200	-0.309	0.107	0.346	0.088
	0.688	0.56	0.57	-0.93	0.100	0.136	-0.218	0.106	0.242	0.098
	0.813	0.35	0.42	-0.93	0.049	0.067	-0.105	0.059	0.110	0.035
	0.938	0.14	0.17	-1.02	0.005	0.009	-0.014	0.010	0.033	0.008
	1.000	0.07	0.09	-0.62	0.001	0.004	-0.003	0.003	0.014	0.013
512	0.563	0.60	1.09	-0.72	0.166	0.403	-0.372	0.085	0.854	0.466
	0.688	0.63	0.88	-0.86	0.153	0.332	-0.388	0.107	0.623	0.305
	0.813	0.57	0.72	-0.87	0.152	0.226	-0.323	0.131	0.285	0.126
	0.938	0.25	0.29	-0.96	0.026	0.036	-0.058	0.028	0.201	0.037
	1.000	0.13	0.15	-0.88	0.007	0.010	-0.015	0.007	0.379	0.098

Table 4: **Head to head** on the same measured curves: RMSE in cosine units over all σ -bins. The spectral column is the family transported through our bridge (Fig. 5) and is δ -inert. Our two columns are the derived small-perturbation form (Eq. 9) and the exact angular link (Eq. 7), both with *zero per-route freedom* on held-out routes: amplitudes and floors come from the two governors of §4.2 fitted on $\{1024 \rightarrow 896, 1024 \rightarrow 512, 1280 \rightarrow 1120\}$. The oracle is a quadratic-in- σ fit on the held-out data itself — the noise ceiling for a 3-dof curve — so beating it on $1024 \rightarrow 768$ means the prediction is at the data’s own resolution.

route	status	spectral	two-term	exact link	oracle
1024→768	held out	0.147	0.093	0.093	0.105
1280→1024	held out	—	0.092	0.049	0.056
1024→512	fit route	0.355	0.093	—	—

the interpolated arm is *worse* than the plain demoted arm through $\sigma \in [0.56, 0.81]$ (paired -0.05 to -0.11), winning only at $\sigma \geq 0.94$; the 768 route’s data term is in any case fatal on its own (+0.09–0.22 over control across the window). On the non-positional residue, the attention-over- N piece has inference-side prior art: attention-entropy corrections derive length-dependent temperatures to preserve the $\log N$ entropy term (Li et al., 2025), and TIDE applies the same dilution correction jointly to resolution and diffusion timestep (Liu et al., 2026) — the closest prior work to a σ -gated resolution correction, though on the inference side; our gate is training-side and set by measured gradient safety, not by an entropy invariance.

The floor as a ledger. Combining the designated probes of §4.2, the floor decomposes additively, each share measured by its own intervention:

$$\text{Floor}_e = \underbrace{\text{reenc}}_{\approx 0 \text{ by control}} + \underbrace{\text{target content}}_{\text{lands } \|\hat{g}_{\text{src}}\|: \text{ angular share } \approx 0} + \underbrace{\text{RoPE}_e}_{\text{erased by PI at } \sigma=1} + \underbrace{\text{Resid}_e}_{\text{remainder}}$$

with totals 0.02–0.04 (896; below the PI probe’s paired resolution, undecomposed), ≈ 0.07 (768; RoPE the large majority, Resid ≈ 0), and ≈ 0.30 (512; RoPE ≈ 0.10 , Resid ≈ 0.20).

F HEAD TO HEAD: PREDICTING HELD-OUT ROUTES

The probes of §4.2 test each account’s *structure*. The remaining test is predictive: fit the reduced account on three routes, then predict two held-out routes from the governors alone. Fit routes: $\{1024 \rightarrow 896, 1024 \rightarrow 512, 1280 \rightarrow 1120\}$; held out: $\{1024 \rightarrow 768, 1280 \rightarrow 1024\}$. Per fit route the excess curve $\overline{\text{gap}}_e(\sigma)$ is fit with a route amplitude and floor on the measured route-uniform $\|\Delta\bar{r}(\sigma)\|$ and each run’s own $\|\bar{g}_{\text{src}}(\sigma)\|$; two governor models ($A(\text{ratio})$ interpolating the fitted amplitudes, and an exponential floor law $F(n) = F_0 e^{-n/\tau}$ in target tokens n) then generate the held-out predictions with zero per-route freedom. Pass criteria were pre-registered in the analysis script before the numbers.

Held-out results, and the comparison. All four pre-registered gates fire (Table 4, Fig. 1). (i) Held-out $1024 \rightarrow 768$: RMSE 0.093, *better than an oracle quadratic fit on the held-out data itself* (0.105), and better than the spectral account both across all bins (0.147) and on the spectral account’s own committed region, where it predicts gap 0 (0.164 against 0.096 for ours). (ii) Held-out $1280 \rightarrow 1024$: RMSE 0.092 under the derived quadratic and 0.049 under the exact link, against oracle 0.056; inside the pre-registered $2\times$ gate either way. (iii) The ratio governor holds at the amplitude level: the two ratio-0.875 routes agree at $z = 0.14$ despite a $1.6\times$ difference in target capacity. (iv) The floor law $F(n) = 0.70 e^{-n/1041 \text{ tok}}$, fit on the 512 and 896 floors, predicts $F(4825) = +0.007$ for the 1120 grid (fitted $+0.002$) and $F(2160) = +0.088$ for the held-out 768 grid, against a measured endpoint floor of $+0.092 \pm 0.012$; a post-hoc leave-896-out re-derivation from 512+1120 alone predicts the 896 floor at $+0.019$ against the measured $+0.019 [+0.010, +0.030]$. The floor law is thus correct on both floors outside its fit set — though its point estimates ride the draw: a full-pipeline bootstrap puts $F(2160)$ at 68% $[+0.042, +0.091]$, so the licensed read is agreement, not exactness.

The margin in Table 4 has a structural reading, not merely a numerical one. The spectral family’s error is not a bad tolerance — δ is inert — but a missing term and a missing governor: it has nothing to put in the high- σ region where the floor lives, and nothing to distinguish $1280 \rightarrow 1024$ from a same-ratio route at a quarter of the token count. Our account’s held-out accuracy comes from exactly the two ingredients it adds.

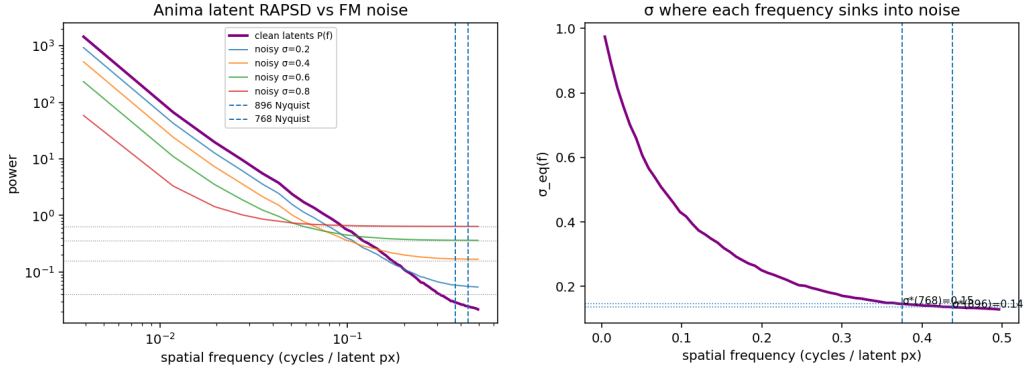
Which functional form, and what the fit does not identify. Refitting the same pipeline under three forms for the data term (a fitted power $A m/G^p$ with p shared and free, the derived quadratic, and the exact link) resolves the form question in the geometry’s favor twice over. The free exponent’s own optimum lands on $p = 2.00$ exactly, recovering the derived quadratic on its own; and the exact link is the best held-out predictor (mean RMSE 0.071 vs 0.093–0.094), with its entire margin coming from $1280 \rightarrow 1024$, where the quadratic systematically *overshoots* the mid- σ peak and the exact link’s saturation removes the overshoot (Appendix Fig. 3). The governors are form-invariant. Two things the fit does *not* establish: the prediction is not within certification resolution anywhere (χ^2/bin 7.6–9.2 held-out under the quadratic), so the licensed claim is shape, magnitude class, and governors at ~ 0.07 – 0.09 RMSE; and the floor law’s functional form is unidentified at our operating points — two anchors and one held-out check cannot separate $\{e^{-n/\tau}, e^{-\sqrt{n}/\ell}, n^{-p}\}$, all of which land within the union of the measured CI and the prediction’s bootstrap spread (Appendix E).

The reduction’s domain: one signed failure, and its mechanism. The 768 route’s mid- σ window is the reduction’s one structural failure, and its shape is diagnostic. Measured excess sits at zero across $\sigma \in [0.56, 0.94]$, *below* the predicted $\text{Floor}_{768} \approx 0.09$ and outside the bootstrap 95% band (Fig. 1): a positive two-term form cannot produce a gap under its own floor, under any coefficients. In the four-term expansion (Eq. 8) this admits exactly two mechanisms, both pre-registered ahead of the probe — either the projected interaction is negative in the window ($I_{768} < 0$, in which case amplitude matching predicts the window center sits where the two perturbation legs have equal orthogonal magnitude, a testable localization), or the graph share is itself σ -dependent, rewriting assumption (iii) rather than (ii). The designated probe is vector-resolved: a demote–re-promote arm reads the shares per (route, bin) from the interventional split $B = \bar{g}_{\text{re-promote}} - \bar{g}_{\text{src}}$, $C = \bar{g}_{\text{demote}} - \bar{g}_{\text{re-promote}}$ (Appendix E, Table 3).

Run, the data picks the first branch with its localization confirmed. $I_{768}(\sigma) < 0$ at every bin (-0.31 at $\sigma = 0.56$, shrinking to -0.014 by 0.94), with the two legs near anti-parallel in-window ($\rho = \cos(B^\perp, C^\perp) \approx -0.93$), and the predicted window center lands at $\sigma \approx 0.69$ ($|B^\perp|/|C^\perp| = 0.98$ there), interior to the measured window. Branch (ii) did not fire: $F_{768}(\sigma)$ decreases monotonically

Table 5: The spectral account (Eq. 4) instantiated on the measured latent spectrum (P at the 896/768/512 cuts = 0.025/0.029/0.065; RAPSD in Fig. 4): predicted safe boundaries t^* per route as the tolerance δ sweeps its range. δ_{reenc} is the measured re-encode noise floor, a zero-free-parameter anchor fixed by our own pipeline. No row reproduces the measured pattern (last line): the δ that matches the safe route’s boundary predicts both failing routes safe by $\sigma = 0.63$, and no δ can produce a floor.

tolerance δ	1024→896	1024→768	1024→512	
$(1 + P_\omega)/2$ (equal power)	0.14	0.15	0.20	classical crossover
0.025	0.50	0.52	0.62	tuned to the measured 896 gate
0.01	0.61	0.63	0.72	default of Xiao et al. (2026)
$\delta_{\text{reenc}} \approx 10^{-5}$	0.98	0.98	0.99	the parameter-free anchor
measured, debiased	≈ 0.5	none certified	unsafe (8/9 bins)	floors .02–.04 / .06–.09 / $\approx .3$



(a) Latent RAPSD against the flow-matching noise floor at four noise levels; dashed lines mark the demoted grids’ Nyquist cuts.

(b) The equal-power slice $\sigma_{\text{eq}}(f)$ of the spectral family. Predicted σ^* : 0.136/0.146/ ~ 0.20 for demotion to 896/768/512.

Figure 4: The spectral account, computed from the measured RAPSD and transported to the training question (§3.2): safety would be predicted from $\sigma \approx 0.14$. Against the measurement (Fig. 2) no tolerance reproduces the pattern (Table 5): the equal-power crossover is off by $\sim 3.5\times$, and the $2\times$ downscale, predicted safe by every tolerance, certifies at no noise level.

to its endpoint value and never drops below it, so the endpoint reads of §4.2 are untouched. The anti-parallel geometry turns out to be universal ($\rho \in [-1.24, -0.62]$ across every route and bin), so what distinguishes routes is amplitude matching, not sign; measured against the exact counterfactual angles, roughly three quarters of the additively-predicted gap is erased by the interference in-window (Appendix E). The window therefore *delimits the reduction’s domain*, and the derived four-term form represents the failure rather than excusing it.

G THE SPECTRAL ACCOUNT: SPECTRUM AND INSTANTIATION

Table 5 evaluates Eq. 4 at the demoted grids’ Nyquist frequencies $f_{\text{cut}} = \frac{1}{2} e/e_0$ on the $N=40$ mean radially-averaged latent power spectrum (64 radial bins; P at the 896/768/512 cuts = 0.025/0.029/0.065; Fig. 4). The equal-power slice $\delta = (1 + P_\omega)/2$ recovers $\sigma_{\text{eq}} = \sqrt{P}/(1 + \sqrt{P})$ exactly. Across $\delta \in [0.001, 0.5]$ the mildest–harshes spread remains below 0.13, and the 1024→896 versus 896→768 boundaries (cut frequencies 0.4375 vs 0.4286) separate by no more than 0.01. The parameter-free anchor δ_{reenc} of §4.2 is measured by the same estimator: RAPSD of the re-encode arm’s latent error (fresh resize–encode minus cached latent, the gradient probe’s own control chain) on the same probe set, interpolated at each route’s cut frequency. Measured ($N=40$, 64 radial bins): error variance 2.7×10^{-5} against latent variance 0.419; $D(f_{\text{cut}}) \approx 1.0 \times 10^{-5}$ at all three cuts (p10–

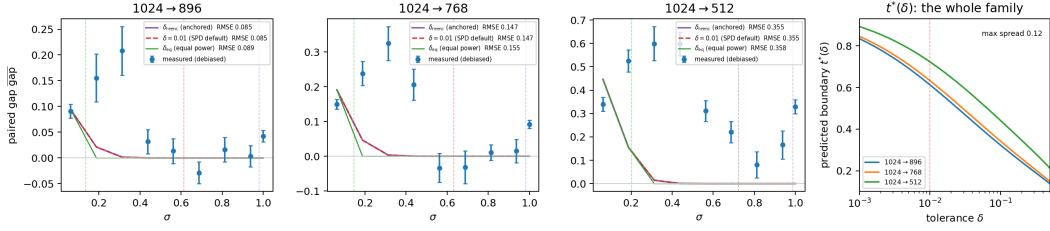


Figure 5: The *whole* tolerance family against the measurement — the detail behind the single red curve of Fig. 1. Left three panels: the measured debiased gap curves against the spectral account transported through our bridge (the diagonal model’s destroyed-band mean-residual mismatch on the measured spectrum, through the measured $\|\bar{g}_{\text{src}}(\sigma)\|$ with a single gain calibrated on the one safe route and no floor — the family expresses none), gated at $t^*(\delta)$ (dashed verticals) for the equal-power slice, SPD’s default $\delta = 0.01$, and the measured anchor δ_{renc} . The transported curve dies by $\sigma \approx 0.35$ on every route, so the two larger gates (0.61–0.72 and 0.98–0.99) act on an already-zero curve and coincide exactly; the equal-power gate (0.14–0.20) is the only one that truncates live curve, and it moves RMSE by at most 0.008. All three therefore land within 0.01 of each other: δ is inert at the curve level. Right: the boundary family $t^*(\delta)$; maximum route spread 0.125 in σ over $\delta \in [10^{-3}, 0.5]$.

p90 over images within $[1, 2] \times 10^{-5}$; per-band SNR $P(f_{\text{cut}})/D(f_{\text{cut}}) \approx 2.5\text{--}4.4 \times 10^3$, giving $t_{\text{renc}}^* = 0.98/0.98/0.99$ for 896/768/512.

The “wrong δ ” objection, closed by construction. One might object that the right δ has simply not been chosen: SPD’s is an inference-side value selected on a speed–quality Pareto ablation over generated images (Xiao et al., 2026), and nothing ties it to gradients. Our setting fixes one *by construction*. The demotion recipe pays a resize–VAE–encode round trip whose latent error has measurable per-band power $D(f)$, expressed in exactly Eq. 4’s units, so $\delta_{\text{renc}}(e) := D(f_{\text{cut}}(e))$ anchors the family with zero free parameters. Measured (above), that anchor is tiny and image-generic, and the boundary moves it forces are measured in Table 5 and Fig. 5: every boundary is pushed to $t^* \approx 0.98$, maximally conservative, and still structurally wrong in both directions at once — it moves all three boundaries together (spread < 0.01), certifies the measured-safe route only above $\sigma \approx 0.98$ where the measurement certifies it from 0.5, and, like every member of the family, predicts the $2\times$ downscale eventually safe when no noise level is. The confrontation of §4.2 therefore does not hinge on the choice of tolerance: the structural failures of §3.2(a–c) are δ -independent at the boundary level, and δ is inert outright at the curve level (Fig. 5).

H CONTROLLED-ADAPTER REPLICATION OF THE VERDICT MAP

The one-adapter limitation of §6 pre-registered a controlled 2×2 factorial: two plain-LoRA checkpoints trained under the shipped probe adapter’s frozen recipe on opposite style clusters (the top and bottom 12 artists by median latent redundancy, hereafter the high-redundancy, visually flat, and low-redundancy, heavily textured, clusters), with stem-level membership manifests frozen before training (124 training images each, identical seeds). Each adapter was probed on 48 stems spanning four membership cells ($N=12$ each): trained-on, held-out images of trained artists, unseen artists of the same cluster, and unseen artists of the opposite cluster, with the cross-cluster stems shared between the two runs so cross-adapter contrasts read paired per stem. Figure 6 repeats the verdict-map recipe per route, overlaying both adapters.

Three reads, stated at the bin-mean level (per-cell tables reproduce from the archived per-image rows). First, the (route, σ) shape replicates on both checkpoints (Fig. 6). Second, the factorial contrasts are null at instrument resolution: the probe-style main effect and the adapter $\times$ probe-style interaction (the pre-registered verdict quantity) are both bounded below the resolvable bin-mean effect (raw-gap paired interaction -0.022 ± 0.027 at 896, -0.015 ± 0.059 at 768), and the membership contrast (trained-on versus held-out) does not replicate in sign across the two adapters. The map is therefore adapter- and content-agnostic on this model at our resolution, supporting a calibrate-once-per-model reading. Third, the one quantity that is *not* checkpoint-invariant is the absolute redraw-floor level:

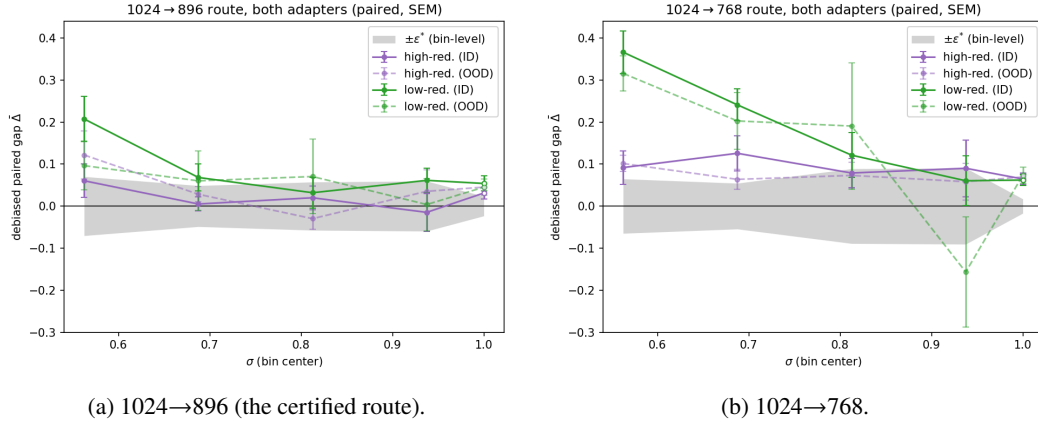


Figure 6: The Fig. 2a recipe repeated on two controlled adapters trained on opposite style clusters, one panel per route ($N=48$ stems per adapter, $D=8/\text{bin}$, high- σ window; same paired debiased estimator and trim; open markers the $\sigma=1$ endpoint bin; axes shared across panels). Unlike the main-text figures, color encodes the *adapter* (high- vs low-redundancy cluster); solid curves are in-distribution stems, the adapter’s own style cluster ($n=36$: trained-on, held-out, and unseen-artist cells), and dashed curves are out-of-distribution stems from the opposite cluster ($n=12$). Gray band: bin-level $\pm\epsilon^*$ per Eq. 11, median of the two runs’ full- N SEMs for the panel’s route. The map replicates: on the certified route all four adapter $\times$ distribution curves reach the band inside the window, on 768 all four sit above it until the top, and within each panel the dashed OOD curves track their solid ID counterparts.

in-window floor cosines sit at ≈ 0.73 for the high-redundancy-cluster adapter versus ≈ 0.50 for the low-redundancy-cluster adapter, uniformly across all four membership cells. The floor’s level is a property of the checkpoint (its gradient stochasticity), while the safety map built on top of it is not.

I PER-BIN VERDICT NUMBERS, AND THE HISTORICAL RECORD

Table 6 is the numerical form of Fig. 2a, the paper’s verdict map, and Table 7 the debiased floor reads that §4.2 scores the graph term on. The remaining tables are the raw-estimator record: retained for continuity and for the reads defended in §4.1 (one-signed bias; paired same-grid contrasts), not cited as debiased evidence. Table 8 is the uncorrected counterpart of the verdict map, and its last two rows are the estimator context plotted in Fig. 2b.

Table 6: **Debiased verdict map** (plotted in Fig. 2a): paired per-image debiased excess $\overline{\text{gap}}$ (SEM) against the re-encoding control, 1024-tier natives ($N=40$, $D=8/\text{bin}$ plus a $\sigma=1$ endpoint bin at the same draws; trimmed per §4.1, retaining 35–40 images per cell). Bin-level ϵ^* at this (N, D) is 0.03–0.08.

σ bin center	1024→896	1024→768	1024→512
0.06	+ .091 (.013)	+ .149 (.014)	+ .340 (.029)
0.19	+ .155 (.047)	+ .238 (.034)	+ .525 (.048)
0.31	+ .208 (.047)	+ .324 (.049)	+ .599 (.073)
0.44	+ .032 (.024)	+ .206 (.045)	+ .598 (.050)
0.56	+ .013 (.024)	− .034 (.042)	+ .312 (.045)
0.69	− .029 (.021)	− .032 (.047)	+ .220 (.046)
0.81	+ .016 (.024)	+ .011 (.022)	+ .081 (.056)
0.94	+ .004 (.021)	+ .015 (.034)	+ .166 (.061)
1.0 (endpoint)	+ .043 (.011)	+ .092 (.012)	+ .329 (.030)

Table 7: **The floor, debiased** (scored in §4.2). **Endpoint:** at $\sigma=1$ the input is exactly ϵ on every grid, so the input-mediated data term vanishes by construction. **x-zero:** the image is removed from input *and* target; any surviving gap is pure graph-shape sensitivity. Debiased draw-limit values $\widehat{\text{gap}}_\infty$ [bootstrap 95% CI over images] from nested draw sweeps (endpoint: $N=12$, $D = 4 \dots 64$; x-zero: $N=40$, $D = 4 \dots 32$). The two columns agree within CI at every route; our pre-registered kill switch (all endpoint gaps ≈ 0) does not fire; and no member of the spectral family can produce a nonzero entry here.

route	endpoint $\widehat{\text{gap}}_\infty$ [95% CI]	x-zero $\widehat{\text{gap}}_\infty$ [95% CI]
native self-floor (cosine)	1.005 [0.994, 1.016]	—
re-encoding control	−0.003 [−0.017, +0.008]	—
1024→896	+0.019 [+0.010, +0.030]	+0.034 [+0.017, +0.058]
1024→768	+0.056 [+0.043, +0.071]	+0.074 [+0.053, +0.094]
1024→512	+0.304 [+0.197, +0.424]	+0.283 [+0.232, +0.332]

Table 8: **Raw estimator** (historical record; superseded by Table 6): demotion gap by σ -bin, 1024-tier natives ($N=40$, bin-mean SEM ~ 0.02 , re-encoding control $|\text{gap}_{\text{reenc}}| \leq 0.054$ everywhere, split-half reliability 0.73–0.83). Bold = within the re-encoding band. Raw values understate mid- σ gaps (the floor attenuates the read) and overstate endpoint floors (single-estimate variance bias); the debiased map corrects both.

σ bin center	0.06	0.19	0.31	0.44	0.56	0.69	0.81	0.94
gap ₈₉₆ (ratio 0.875)	.110	.162	.148	.137	.048	.030	.030	.053
gap ₇₆₈ (ratio 0.75)	.144	.216	.208	.223	.164	.163	.115	.063
gap ₅₁₂ (ratio 0.5)	.348	.410	.355	.469	.430	.391	.289	.296
cos _{floor}	.84	.65	.51	.65	.70	.77	.80	.83
$\ g\ $ (native)	2.6	0.5	0.4	0.5	0.7	1.1	2.0	9.3

Table 9: Raw endpoint reads ($D=16$, $N=40$) beside their debiased draw-limit values (Table 7), the debiasing’s revision record: the 768 floor halves, the 896 floor surfaces.

route	raw endpoint ($D=16$)	endpoint $\widehat{\text{gap}}_\infty$ [95% CI]
native self-floor (cosine)	0.85	1.005 [0.994, 1.016]
re-encoding control	−0.038	−0.003 [−0.017, +0.008]
1024→896	−0.009	+0.019 [+0.010, +0.030]
1024→768	+0.127	+0.056 [+0.043, +0.071]
1024→512	+0.326	+0.304 [+0.197, +0.424]

Table 10: Iso-severity probe (**raw estimator**): gap by σ -bin on the 1280-tier cache ($N=24$, 4 draws/bin) against the corpus 1024→896 curve. The ratio-matched routes coincide within ~ 1.4 combined SEM at every bin despite a $1.6\times$ difference in absolute target size; the same-native harsher-ratio route separates. *896’s canonical 16-draw endpoint is −0.009 raw, +0.019 debiased (Table 7).

σ	0.125	0.375	0.625	0.875	1.0
re-encoding control	+0.047	−.005	+0.048	+0.020	−.012
1280→1120 (ratio .875)	.129	.110	.054	−.012	−.007
1280→1024 (ratio .80)	.170	.178	.111	.002	.061
1024→896 (ratio .875)	.132	.076	.077	.049	.100*

Table 11: Ratio-transfer run (**raw estimator**): 896-tier natives ($N=40$), pre-registered bar “gap₇₆₈ within the re-encoding band at $\sigma \geq 0.5$ ”: **fails** (residual 0.06–0.12, $\sim 2\times$ the 1024→896 plateau; $\sim$ half of it is expected estimator bias, §4.1; the separation is independently confirmed by the debiased same-grid floors, §4.2).

σ bin	0.06	0.19	0.31	0.44	0.56	0.69	0.81	0.94
896→768	.114	.209	.168	.143	.124	.056	.092	.061
896→512	.318	.372	.308	.340	.320	.280	.329	.217

Table 12: Route-uniformity (in amplitude) of the mean prediction residual: cross-grid excess of $\bar{r} = \mathbb{E}_\epsilon [\hat{v} - (\epsilon - x)]$ over split-half floors (relative L_2 ; same-grid re-encoding control ≤ 0.02 everywhere). The three main routes coincide within $\sim \pm 0.02$ at every σ while their gradient floors span 0 to 0.3. This is a norm read; the direction-resolved follow-up (Appendix E) shows the underlying mismatch directions are image- and route-specific.

σ	0.125	0.375	0.625	0.875	1.0
1280→1024	.885	.825	.715	.465	.360
1024→896	.862	.808	.706	.459	.378
896→768	.860	.814	.715	.466	.393
768→512	.897	.872	.767	.508	.435

Table 13: Depth localization of the floor (**raw estimator**): mean per-block gap at $\sigma=1$, early = blocks 0–9, late = 14–27, for the endpoint (ep) and x-zero (xz) probes. Early blocks carry $\sim 3\times$ the late-block gap. The apparent content share (ep – xz) is a late-block minority effect in raw units; the debiased draw-limit comparison finds ep = xz within CI at the whole-adapter level (Table 7). Within a block every module type sits in 0.22–0.31 at 512/ $\sigma=1$, including the query and key rows of the fused QKV projection — landing-side uniformity, which is why the positional share needed an origin-side intervention to find.

route	ep early	ep late	xz early	xz late
1024→512	.357	.223	.351	.125
1024→768	.164	.085	.121	.033
1024→896	.023	.015	.044	.012

Table 14: Positional-interpolation intervention at the $\sigma=1$ endpoint ($N=40$, 16 draws; **raw** arm values; the paired Δ column is a same-grid contrast in which the finite-draw bias cancels to first order): exact phase-geometry alignment erases the mild-route floor and $\sim 30\%$ of the harsh-route floor.

route	plain (SEM)	PI-aligned (SEM)	paired Δ (SEM)	improved
1024→896 (control)	−0.021 (.041)	−0.040 (.040)	—	—
1024→768	+0.080 (.048)	−0.001 (.039)	+0.081 (.031)	78%
1024→512	+0.320 (.058)	+0.224 (.056)	+0.096 (.039)	70%